

A PROJECT REPORT ON
**HOMOMORPHIC ENCRYPTION BASED VIDEO COPY DETECTION
IN MULTI VIEW VIDEOS STORED IN A CLOUD**

Submitted By

SK. SHARIF **22JD1A05A6**

N. MALLIKA **22JD1A0575**

N. DURGA TEJESH **22JD1A0578**

in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE & ENGINEERING

Under the Esteemed guidance of

Mr. G. PRANITH B.Tech,M.Tech

(Assistant Professor)



**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
ELURU COLLEGE OF ENGINEERING AND TECHNOLOGY
DUGGIRALA (V), PEDAVEGI (M), ELURU-534004
(AFFILIATED TO JNTUK, KAKINADA, ANDHRA PRADESH (INDIA))
2022-2026**

DECLARATION

We here by declaring that the project entitled “**HOMOMORPHIC ENCRYPTION BASED VIDEO COPY DETECTION IN MULTI VIEW VIDEOS STORED IN A CLOUD**”submit for the B.Tech. (CSE) degree is our original work and the project has not formed the basis for the award of any other degree, diploma, fellowship or any other similar titles.

SK. SHARIF (22JD1A05A6)

N. MALLIKA (22JD1A0575)

N. DURGA TEJESH (22JD1A0578)

Place :

Date :

CERTIFICATE

This is to certify that the Project titled “**HOMOMORPHIC ENCRYPTION BASED VIDEO COPY DETECTION IN MULTI VIEW VIDEOS STORED IN A CLOUD**” is the bonafide work carried out by, **SK. SHARIF (22JD1A05A6)**, **N. MALLIKA (22JD1A0575)**, **N. DURGA TEJESH (22JD1A0578)** students of B.Tech (CSE) of Eluru College of Engineering and Technology, affiliated to JNT University, Kakinada, AP (India) during the academic year 2025-26, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (Computer Science and Engineering) and that the project has not formed the basis for the award previously of any other degree, diploma, fellowship or any other similar title.

Signature of the Guide

Mr. G PRANITH, B. Tech, M. Tech

Assistant professor

Signature of the HOD

Dr. S. SURESH, B.Tech, M.Tech, Ph.D

Professor

EXTERNAL EXAMINER

VISION – MISSION - PEOs

Institute Vision	To Achieve Excellence in Engineering Education
Institute Mission	IM1: To deliver quality education through good infrastructure, facilities and committed staff. IM2: To train students as proficient, competent and socially responsible engineers. IM3: To promote research and development activities among faculty and students for betterment of the society.
Department Vision	Empower the students of computer science and engineering department to be technologically strong, innovative and global citizens maintaining human values.
Department Mission	DM 1: Inspire students to become self-motivated and problem solving individuals. DM 2: Furnish students for professional career with academic excellence and leadership skills. DM 3: Create centre of excellence in Computer Science and Engineering. DM 4: Empower the youth and rural communities with computer education.
Program Educational Objectives (PEOs)	Graduates of Computer Science & Engineering will: PEO1: Excel in Professional career through knowledge in mathematics and engineering principles. PEO2: Able to pursue higher education and research. PEO3: Communicate effectively, recognize, and incorporate societal needs in their professional endeavors. PEO4: Adapt to technological advancements by continuous learning.

PROGRAM OUTCOMES

1	Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
2	Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations
4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7	Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9	Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES

PSO1: Have proficiency in algorithms, networking, s/w Engineering and Web Applications for systematic design of computer based system of varying complexity.

PSO2: Qualifying national, international level competitive examinations for successful higher studies and employment.

PROJECT MAPPINGS

Batch No:	B - 2
Project Title	Homomorphic Encryption Based Video Copy Detection in Multi View Videos Stored in a Cloud
Project Domain	Machine Learning
Type of the Project	Research – oriented
Guide Name	G. PRANITH,B.Tech, M. Tech
Student Roll No	Student Name
22JD1A05A6	SK. Sharif
22JD1A0575	N. Mallika
22JD1A0578	N. Durga Tejesh

COURSE OUTCOMES : At the end of the Course/Subject, the students will be able to

CO.No	Course Outcomes (COs)	POs	PSOs	Blooms Taxonomy & Level
R20C501.1	Identify the real world problem with a set of requirements to design a solution.	1,2,3,4,5,6,9,10,11,12	1,2	Level-2: Applying
R20C501.2	Implement, Test and Validate the solution against the requirements for a given problem.	1,2,3,4,5,7,8,9,10,11,12	1,2	Level-3: Analysing
R20C501.3	Lead a team as a responsible member in developing software solutions for real world problems and societal issues with ethics.	1,3,4,5,6,7,8,9,10,12	1,2	Level-4: Analysing
R20C501.4	Participate in discussions to bring technical and behavioral ideas for good solutions.	1,2,3,4,5,6,7,8,9,10,11, 12	1,2	Level-5: Evaluating
R20C501.5	Express ideas with good communication skills during presentations.	1,5,6,7,8,9,10,12	1,2	Level-1: Creating
R20C501.6	Learn new technologies to contribute in the software industry for optimal solutions.	1,2,3,4,5,7,8,9,10,11,12	1,2	Level-6: Creating

Course Outcomes vs PO's Mapping:

Courses Out Comes	P01	P02	P03	P04	P05	P06	P07	P08	P09	P10	P011	P012
R20C501.1	3	3	3	2	3	3	-	3	3	3	3	3
R20C501.2	3	3	3	3	3	-	3	3	3	3	3	3
R20C501.3	2	-	3	3	3	3	3	3	3	3	3	3
R20C501.4	3	3	3	3	2	3	3	3	3	3	3	3
R20C501.5	3	-	-	-	3	3	3	3	3	3	3	3
R20C501.6	2	3	3	3	3	-	3	3	3	3	3	3
Total	16	12	15	14	17	12	15	18	18	18	18	18
Average	2.6	2	2.6	2.3	2.8	2	3	3	3	3	3	3

Justification of Mapping of Course Outcomes with Program Outcomes: POs mapped with CO.No:R20C501.1:**PO-1: Engineering Knowledge:**

Applying the various concepts learned from engineering to gather requirements.

PO-2: Problem Analysis:

Identify the formulae and obtaining the layout of the design using mathematical principles.

PO-3: Design/Development of solutions:

Identifying and Designing solutions for the problems related to health, cultural, society and environmental considerations.

PO-4: Conduct investigation of complex problems:

Researching various journals and published papers and knowing how to solve complex problems.

PO-5: Modern Tool Usage:

Selecting and Implementing the latest versions of the applications for modelling the complexity of the project.

PO-6: The Engineer and Society:

Implementing this knowledge to assess health and safety considerations.

PO-8: Ethics:

Implementing professional ethics and norms in engineering practices.

PO-9: Individual and Teamwork:

Individuals researched on the interested domain and concluded the requirements as a team.

PO-10: Communication:

Communication is needed to identify the problem and work as a team to define a solution.

PO-11: Project Management and Finance:

Implementing the engineering principles to work on complex problems.

PO-12: Life-long Learning:

Recognize the importance of technical knowledge in future studies

POs mapped with CO.No:R20C501.2:**PO-1: Engineering Knowledge:**

Implementing the various concepts learned from engineering to gather requirements and design and test a solution.

PO-2: Problem Analysis:

Identify the formulae and obtaining the layout of the design and validate using mathematical principles.

PO-3: Design/Development of solutions:

Identifying and Designing solutions for the problems related to health, cultural, society and environmental considerations.

PO-4: Conduct investigation of complex problems:

Researching various journals and published papers and knowing how to solve complex problems.

PO-5: Modern Tool Usage:

Selecting and Implementing the latest versions of the applications for modelling the complexity of the project.

PO-7: Environment and Sustainability

Implementing this knowledge to assess its affects towards environment.

PO-8: Ethics:

Implementing professional ethics and norms in engineering practices.

PO-9: Individual and Teamwork:

Individuals researched on the interested domain and concluded the requirements as a team.

PO-10: Communication:

Communication is needed to Recognize the problem and work as a team to define a solution.

PO-11: Project Management and Finance:

Implementing the engineering principles to work on complex problems.

PO-12: Life-long Learning:

Recognize the importance of technical knowledge in future studies.

POs mapped with CO.No:R20C501.3:**PO-1: Engineering Knowledge:**

Implementing the various concepts learned from engineering to gather requirement and work as a team to design and test a solution.

PO-3: Design/Development of solutions:

Recognizing and Designing solutions for the problems related to health, cultural, society and environmental considerations.

PO-4: Conduct investigation of complex problems:

Researching various journals and published papers and knowing how to solve complex problems.

PO-5: Modern Tool Usage:

Selecting and Implementing the latest versions of the applications for modelling the complexity of the project.

PO-6: The Engineer and Society:

Implementing this knowledge to assess health and safety considerations.

PO-7: Environment and Sustainability

Implementing this knowledge to assess its affects towards environment.

PO-8: Ethics:

Implementing professional ethics and norms in engineering practices.

PO-9: Individual and Teamwork:

Individuals researched on the interested domain and concluded the project as a team.

PO-10: Communication:

Communication is needed to Recognize the problem and work as a team to obtain a solution.

PO-12: Life-long Learning:

Recognize the importance of technical knowledge in future studies.

POs mapped with CO.No:R20C501.4:**PO-1: Engineering Knowledge:**

Implementing the various concepts learned from engineering to gather requirement and work as a team to design and test a solution.

PO-2: Problem Analysis:

Recognize the formulae and obtaining the layout of the design and validate using mathematical principles.

PO-3: Design/Development of solutions:

Recognizing and Designing solutions for the problems related to health, cultural, society and environmental considerations.

PO-4: Conduct investigation of complex problems:

Researching various journals and published papers and knowing how to solve complex problems.

PO-5: Modern Tool Usage:

Selecting and Implementing the latest versions of the applications for modelling the complexity of the project.

PO-6: The Engineer and Society:

Implementing this knowledge to assess health and safety considerations.

PO-7: Environment and Sustainability

Implementing this knowledge to assess its affects towards environment.

PO-8: Ethics:

Implementing professional ethics and norms in engineering practices. PO-9:

Individual and Teamwork:

Individuals researched on the interested domain and concluded the project as a team.

PO-10: Communication:

Communication is needed to Recognize the problem and work as a team to obtain a solution.

PO-11: Project Management and Finance:

Implementing the engineering principles to work on complex problems.

PO-12: Life-long Learning:

Recognize the importance of technical knowledge in future studies.

POs mapped with CO.No:R20C501.5:**PO-1: Engineering Knowledge:**

Implementing the various concepts learned from engineering to gather requirement and work as a team to design and test a solution.

PO-5: Modern Tool Usage:

Selecting and Implementing the latest versions of the applications for modelling the complexity of the project.

PO-6: The Engineer and Society:

Implementing this knowledge to assess health and safety considerations.

PO-7: Environment and Sustainability

Implementing this knowledge to assess its affects towards environment.

PO-8: Ethics:

Implementing professional ethics and norms in engineering practices.

PO-9: Individual and Teamwork:

Individuals researched on the interested domain and concluded the project as a team.

PO-10: Communication:

Communication is needed to Recognize the problem and work as a team to obtain a solution.

PO-12: Life-long Learning:

Recognize the importance of technical knowledge in future studies.

POs mapped with CO.No:R20C501.6:**PO-1: Engineering Knowledge:**

Implementing the various concepts learned from engineering to gather requirement and work as a team to design and test a solution.

PO-2: Problem Analysis:

Recognize the formulae and obtaining the layout of the design and validate using mathematical principles.

PO-3: Design/Development of solutions:

Identifying and Designing solutions for the problems related to health, cultural, society and environmental considerations.

PO-4: Conduct investigation of complex problems:

Researching various journals and published papers and knowing how to solve complex problems.

PO-5: Modern Tool Usage:

Selecting and Implementing the latest versions of the applications for modelling the complexity of the project.

PO-7: Environment and Sustainability

Implementing this knowledge to assess its affects towards environment.

PO-8: Ethics:

Implementing professional ethics and norms in engineering practices.

PO-9: Individual and Teamwork:

Individuals researched on the interested domain and concluded the project as a team.

PO-10: Communication:

Communication is needed to Recognize the problem and work as a team to obtain a solution.

PO-11: Project Management and Finance:

Implementing the engineering principles to work on complex problems.

PO-12: Life-long Learning:

Recognize the importance of technical knowledge in future studies.

Course Outcomes vs PSOs Mapping:

Courses Out Comes	PSO1	PSO2
R20C501.1	3	3
R20C501.2	3	3
R20C501.3	3	3
R20C501.4	3	3
R20C501.5	3	3
R20C501.6	3	3
Total	18	18
Average	3	3

Justification of Mapping of Course Outcomes with Program Specific Outcomes:**PSOs mapped with CO.No:R20C501.1:**

PSO-1: Gaining knowledge in software engineering practices to gather requirements.

PSO-2: Able to use this knowledge in further studies.

PSOs mapped with CO.No:R20C501.2:

PSO-1: Gaining knowledge in software engineering practices like algorithms to design a solution.

PSO-2: Able to use this knowledge in further studies.

PSOs mapped with CO.No:R20C501.3:

PSO-1: Gaining knowledge in software engineering practices to gather requirements and networking for designing and communication.

PSO-2: Able to use this knowledge in further studies.

PSOs mapped with CO.No:R20C501.4:

PSO-1: Gaining knowledge in software engineering practices to gather requirements and bring in technical and behavioural ideas.

PSO-2: Able to use this knowledge in further studies.

PSOs mapped with CO.No:R20C501.5:

PS0-1: Gaining knowledge in software engineering practices to gather requirements and using good communication skills to express ideas.

PSO-2: Able to use this knowledge in further studies.

PSOs mapped with CO.No:R20C501.6:

PS0-1:Gaining knowledge in software engineering practices to learn to explore new technologies.

PSO-2: Able to use this knowledge in further studies.

Mapping Level	Mapping Description
1	Low Level Mapping with PO & PSO
2	Moderate Mapping with PO & PSO
3	High Level Mapping with PO & PSO

Member 1: SK. SHARIF (22JD1A05A6)

Member 2: N. MALLIKA (22JD1A0575)

Member 3: N. DURGA TEJESH (22JD1A0578)

PROJECT GUIDE

ACKNOWLEDGEMENT

First, we would like to express our deepest appreciation to our advisor and guide, Mr. G Pranith B.Tech,M.Tech for his sincere guidance, heart full encouragement, and continuous support during our B. Tech project at Eluru College of Engineering and Technology. His vision for insights into the problem resolution has greatly inspired us to put things in the right direction. It has been a great pleasure to conduct our project under his guidance.

We must also express our sincere thanks to our B.tech Project Co-Ordinator and Assoc. Professor Mrs.K V.JHANSI RANI B.Tech,Mtech,Ph.D, The Head of the Department Dr. S. SURESH B.Tech,M.Tech,Ph.D, and to all of the faculty members of the Computer Science and Engineering of Eluru College of Engineering and Technology for their endeavors in our academic education.

We would like to express our gratitude to Dr. P.BALA KRISHNA PRASAD , Principal, and the Management of ELURU COLLEGE OF ENGINEERING & TECHNOLOGY for motivating and providing us adequate facilities and continuous improvements and suggestion without which this project would not have seen the light of the day.

SK.SHARIF (22JD1A05A6)

N.MALLIKA (22JD1A0575)

N.DURGA TEJESH (22JD1A0578)

ABSTRACT

With the exponential growth of video content shared and stored on cloud platforms, securing copyrighted videos and detecting unauthorized copies has become crucial. Multi-view videos, captured from multiple angles simultaneously, add complexity to this challenge due to large data volumes and varied perspectives. Traditional copy detection methods often fail to preserve privacy, especially when outsourced to untrusted cloud environments. To address this, we propose a homomorphic encryption-based video copy detection system that ensures privacy-preserving processing and efficient detection. Homomorphic encryption allows operations on encrypted data, enabling the cloud to perform similarity comparisons without accessing the original video content. This approach maintains the confidentiality of the videos while enabling robust and accurate detection of copied content across different views. The proposed system enhances both security and effectiveness in identifying video copies in a cloud-based environment.

LIST OF FIGURES

Figure No.	Name of the Figure	Page No.
4.1.1	System Architecture	18
4.2.1	Use Case Representation	19
4.2.2	Sequence Diagram	19
4.2.3	Activity Diagram	20
4.2.4	Class Diagram	21
4.2.5	Component Diagram	21
4.2.6	Collaboration Diagram	22
4.2.7	Deployment Diagram	22
7.1	Home page	57
7.2	New User Sign up	57
7.3	Entering sign up details	58
7.4	User Login	58
7.5	Welcome Page After Login	59
7.6	Upload & Detect Video Copy	59

7.7	Uploading video	60
7.8	Unique or Copy	60
7.9	Output	61
7.10	Checking & Downloading Video	61

TABLE OF CONTENTS

S.NO	TITLE	PAGE NO
1.	INTRODUCTION	1
	1.1 Motivation	2
	1.2 Problem Definition	2
	1.3 Objective of the Project	2
2.	LITERATURE SURVEY	4
3.	SYSTEM ANALYSIS	6
	3.1 Existing System	6
	3.1.1 Disadvantages of Existing System	6
	3.2 Proposed System	8
	3.2.1 Advantages of Proposed System	9
	3.3 System Requirements	10
	3.3.1 Hardware Requirements	10
	3.3.2 Software Requirements	10
	3.4 Functional Requirements	10
	3.5 Non-Functional Requirements	10
	3.5.1 Advantages	11
	3.5.2 Disadvantages	12
	3.5.3 Key Learning	12
	3.6 System Study	12
	3.6.1 Study of the System	13
	3.7 Modules	16
4	SYSTEM DESIGN	18
	4.5 System Architecture	18
	4.6 UML Representation	19
	4.7 Input & Output Design	23
5	SYSTEM IMPLEMENTATION	25
	5.1 Source Code	25
	5.2 Algorithms	40

5.3	Software Environment	42
6.	TESTING	52
6.1	Types of Testing	52
6.2	Test strategy and approach	53
7.	RESULTS	57
8.	CONCLUSION & FUTURE WORK	62
9.	REFERENCES	64

1. INTRODUCTION

The widespread use of cloud services and the exponential growth in multimedia content creation have revolutionized how videos are stored, shared, and consumed globally. With increasing internet penetration and the emergence of advanced camera technologies, multi-view videos—videos captured simultaneously from different angles—are becoming more prevalent in domains such as surveillance, virtual reality, telemedicine, and entertainment. While these developments offer enhanced user experiences and broader application scopes, they also raise serious concerns regarding unauthorized duplication, content piracy, and intellectual property violations. Traditional video copy detection systems operate on plaintext data and often require centralized control, posing significant risks to data confidentiality, especially when the content is stored or processed in the cloud. These systems fall short in scenarios where privacy is paramount, such as confidential surveillance footage or proprietary visual media. Moreover, conventional algorithms may not perform well with multi-view videos, as they struggle to maintain robustness against viewpoint changes, occlusions, and transformations. This underscores the necessity for a privacy-preserving, scalable, and robust solution capable of effectively detecting video copies across varying views.

Homomorphic encryption (HE) has emerged as a powerful tool in the field of privacy-preserving computing. It enables computations to be performed directly on encrypted data, producing encrypted results that, when decrypted, match the output of operations carried out on the original plaintext. Leveraging this property, this project aims to develop a homomorphic encryption-based framework for secure video copy detection. By encrypting the extracted video features before uploading them to the cloud, the proposed system ensures that the underlying data remains private, even during search and comparison operations.

The system first extracts view-invariant features from each frame of a multi-view video using advanced computer vision techniques. These features, designed to be resistant to changes in perspective and scale, serve as fingerprints for the video content. Once encrypted using a suitable HE scheme, the fingerprints are stored securely in a cloud database. During detection, the system compares encrypted query features against stored encrypted fingerprints using homomorphic operations, thus identifying potential copies without compromising the confidentiality of either the query or the stored videos.

The integration of homomorphic encryption with multi-view video analysis addresses a critical gap in current research by combining data privacy with effective copy detection. This novel approach not only ensures security and trust in cloud-based video storage systems but also opens up new possibilities for secure media content management in sensitive environments. Through this project, we aim to advance the state of the art in both multimedia security and privacy-preserving computing.

1.1 Motivation:

The rapid growth of cloud-based video platforms such as YouTube and Netflix has led to an explosion in multimedia content, making the protection of copyrighted videos increasingly important. Unauthorized copying and redistribution not only violate intellectual property rights but also cause significant losses to content creators. This problem becomes more challenging with multi-view videos, where content is captured from multiple angles, increasing data volume and complexity. Traditional copy detection methods often require access to raw video data, raising serious privacy concerns, especially in untrusted cloud environments. To address these issues, this project is motivated by the need for a secure and efficient solution that can detect video copies without exposing sensitive data, using Homomorphic Encryption to enable processing on encrypted content while preserving confidentiality and ensuring accurate detection.

1.2 Problem Definition:

The increasing use of cloud platforms for storing and sharing large-scale video content, especially multi-view videos, has made detecting unauthorized copies a critical yet challenging task. Existing video copy detection methods are often inefficient in handling the high data volume and variations in perspectives, and they typically require access to unencrypted video data, leading to serious privacy risks in untrusted cloud environments. This creates a conflict between the need for accurate copy detection and the requirement to preserve content confidentiality. Therefore, there is a need to develop a secure and efficient system that can detect copied videos across multiple views while ensuring data privacy, which can be achieved by leveraging techniques like Homomorphic Encryption to enable processing directly on encrypted data without exposing the original content.

1.3 Objective of the Project:

The main objective of this project is to develop a smart system that:

- To design a secure video copy detection system for cloud-based environments that can handle large-scale and multi-view video data efficiently.
- To develop a privacy-preserving framework using Homomorphic Encryption that enables processing and comparison of videos without revealing the original content.
- To accurately detect unauthorized copies of videos even when they are captured from different angles or perspectives.
- To reduce the risk of data leakage and ensure confidentiality when outsourcing video

processing to untrusted cloud servers.

- To create a scalable solution that can adapt to the growing volume of multimedia content in modern cloud platforms.
- To improve the efficiency and reliability of traditional video copy detection methods by incorporating encrypted-domain operations.

2. LITERATURE SURVEY

The paper “*A Secure and Privacy-Preserving Video Surveillance System Using Homomorphic Encryption*” by Zhang, H., Wang, C., and Yu, S. presents an advanced framework designed to ensure secure video surveillance in cloud environments. The authors utilize Fully Homomorphic Encryption to enable computations such as motion detection and object tracking directly on encrypted video streams. This approach eliminates the need to decrypt sensitive data, thereby maintaining strict privacy and security. The study demonstrates that encrypted-domain processing is feasible and effective, especially in scenarios where cloud servers cannot be fully trusted. It also highlights the growing importance of privacy-preserving techniques in modern surveillance systems.

In the paper “*Robust Video Copy Detection Using Spatio-Temporal Features*”, Chen, L., Wu, F., and Li, H. propose a powerful video fingerprinting method based on spatio-temporal interest points. This technique captures both spatial and temporal characteristics of video content, making it highly robust against common transformations such as scaling, rotation, compression, and format conversion. The method improves the reliability of detecting copied or manipulated videos and provides a strong foundation for handling complex scenarios such as multi-view video systems. Their work significantly contributes to the advancement of feature-based video copy detection.

The research work “*Multi-view Video Summarization and Matching for Surveillance Applications*” by Liu, X., Zhang, W., and Xu, C. focuses on analyzing and managing multi-view video data. The authors propose methods for summarizing video content captured from multiple cameras and identifying overlapping or redundant information across different views. This is particularly useful in surveillance applications where multiple cameras capture the same scene from different angles. The study highlights the computational complexity and challenges involved in matching multi-view video content and emphasizes the need for efficient algorithms.

Furthermore, the paper “*Privacy-Preserving Multimedia Content Analysis: Opportunities and Challenges*” by Sadeghi, A., Schneider, T., and Winandy, M. provides a comprehensive survey of existing techniques for secure multimedia processing. The authors discuss various encryption-based approaches and identify Homomorphic Encryption as one of the most promising solutions for enabling computations on encrypted data. They also analyze the trade-offs between security, computational cost, and system performance, offering valuable insights into the practical challenges of implementing privacy-preserving systems.

In “*Homomorphic Encryption for Cloud-Based Video Processing*”, Bianchi, T., Piva, A., and Barni, M. explore the application of encryption techniques in cloud-based video processing systems. Instead of encrypting raw video data, the authors focus on encrypting video descriptors, which significantly reduces computational overhead while still preserving privacy. Their work demonstrates that encrypted-domain processing can be efficiently applied to tasks such as content-based video retrieval, making it highly relevant for real-world applications.

Finally, the paper “*Secure Outsourced Media Search with Encrypted Domain Content Matching*” by Song, Y., and Zhou, Y. introduces a secure framework for searching and matching multimedia content stored in encrypted form. The system uses advanced encryption methods to allow query processing without revealing the actual content. This approach is particularly useful for cloud storage systems where privacy is a major concern. The study shows that secure and efficient media search is achievable even in untrusted environments.

Overall, these studies collectively highlight the importance of combining robust video analysis techniques with strong encryption methods to address the challenges of privacy, security, and efficiency in modern cloud-based video systems.

3. SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

In the current digital landscape, various systems have been developed to detect unauthorized video copies. These systems primarily rely on content-based video retrieval (CBVR) techniques and digital video fingerprinting. The typical workflow involves extracting unique, representative features from video files and comparing them to features stored in a database to identify duplicates or near-duplicates. While effective in controlled scenarios, existing systems exhibit significant limitations when applied to modern, complex environments such as multi-view video repositories in the cloud.

Most existing video copy detection techniques operate on plaintext data, which poses serious privacy and security concerns when sensitive or proprietary videos are involved. These systems require full access to the video content, making them unsuitable for cloud-based applications where user data privacy is critical. In such scenarios, the risk of data leakage or misuse is high, especially if cloud service providers or third-party systems are compromised.

Additionally, current systems are not well-equipped to handle multi-view videos. These videos, captured from multiple angles, introduce high variability in appearance due to changes in perspective, occlusion, lighting, and camera motion. Traditional copy detection algorithms often fail to maintain robustness against such transformations, leading to decreased accuracy and higher false positive/negative rates. They are typically designed for single-view, linear video streams, making them ineffective for spatially and temporally diverse datasets.

Another limitation of existing systems is their lack of scalability and efficiency in large-scale environments. With the rapid growth of multimedia data in cloud repositories, performance bottlenecks arise due to the computational complexity of feature extraction, storage, and comparison. These systems also lack built-in mechanisms for encrypted domain operations, meaning they cannot operate without exposing sensitive data to the system or administrators.

In summary, the existing systems fall short in three critical areas: **privacy preservation, multi-view video robustness, and secure cloud integration**. These limitations highlight the need for a new approach—one that can provide secure, accurate, and efficient video copy detection without compromising user data privacy. This forms the basis for the proposed system, which leverages **homomorphic encryption and view-invariant feature extraction techniques** to overcome the deficiencies of current solutions.

3.1.1 DISADVANTAGES OF EXISTING SYSTEM

1. Lack of Privacy Preservation

- Most existing systems process video content in plaintext form, making sensitive data vulnerable to unauthorized access or misuse during storage, transmission, or analysis.
- These systems cannot be trusted in cloud environments where third-party services may have access to unencrypted user content.

2. Incompatibility with Multi-View Videos

- Existing copy detection algorithms are typically optimized for single-view videos and fail to accurately detect duplicates when the content is captured from different camera angles.
- Viewpoint variations, occlusions, and lighting changes in multi-view setups significantly reduce detection accuracy.

3. Poor Scalability in Cloud Environments

- Many systems are not designed for scalable processing in distributed cloud storage, leading to performance bottlenecks with large datasets.
- The computational cost of comparing video fingerprints across massive repositories becomes prohibitive without cloud-optimized algorithms.

4. High False Positives/Negatives

- Traditional feature extraction methods often lack robustness against common video transformations such as cropping, compression, re-encoding, and partial modifications.
- This results in an increased number of incorrect detections, undermining system reliability.

5. No Support for Encrypted Domain Operations

- Existing methods require decryption of video data or features before processing, violating the principle of secure computation.
- They do not leverage modern cryptographic techniques such as homomorphic encryption to enable secure computation on encrypted data.

6. Limited Real-Time Capability

- Due to their computational inefficiencies and heavy reliance on sequential processing, many existing systems are unsuitable for real-time or near-real-time video monitoring and copy detection tasks.

7. Centralized Architecture Risks

- Systems that rely on centralized databases or processing servers present single points of failure and increase the risk of data breaches or service outages.

3.2 PROPOSED SYSTEM

To overcome the limitations of existing video copy detection systems, this project proposes a **Homomorphic Encryption Based Video Copy Detection System** specifically designed for **multi-view videos stored in a cloud environment**. The goal is to provide a **secure, privacy-preserving, and efficient framework** that can detect duplicate or pirated video content without exposing the original data to unauthorized parties, including cloud service providers. The core innovation of the proposed system lies in the use of **homomorphic encryption (HE)**, which enables computations to be carried out on encrypted data. This ensures that sensitive video content remains confidential throughout the feature extraction, storage, and comparison processes. Even the copy detection operations themselves are performed directly in the encrypted domain, eliminating the risk of data exposure.

The proposed system operates in the following key stages:

1. **View-Invariant Feature Extraction**

Multi-view videos are first processed to extract **robust, view-invariant feature vectors** using advanced computer vision techniques such as 3D convolutional neural networks (3D-CNNs) or spatio-temporal descriptors. These features are designed to capture the semantic content of videos regardless of the camera angle or viewpoint, ensuring reliable comparison across views.

2. **Homomorphic Encryption of Features**

The extracted feature vectors are then encrypted using a **homomorphic encryption scheme** (e.g., BFV, CKKS, or Paillier). This step transforms the features into a secure format that supports mathematical operations such as addition and multiplication while still encrypted. This guarantees that the system never has access to the raw data or unencrypted features.

3. **Secure Cloud Storage and Indexing**

Encrypted features are uploaded and indexed in a **cloud-based database**, which is optimized for secure and efficient retrieval. The system architecture supports distributed storage and horizontal scaling, allowing it to manage large volumes of video content without performance degradation.

4. **Encrypted-Domain Copy Detection**

When a new video (or a query) is introduced, its features are extracted and encrypted in the same manner. These encrypted query features are then compared against the encrypted features in the cloud database using **homomorphic operations**, enabling secure matching without ever decrypting the data.

5. **Copy Detection Output and Alerts**

If a matching encrypted signature is found—indicating potential duplication or piracy—the system flags it and generates a secure report. Optionally, alerts can be sent to content owners or administrators for further action.

3.2.1 ADVANTAGES OF PROPOSED SYSTEM

1. End-to-End Privacy Preservation

- The use of **homomorphic encryption** ensures that video features are never exposed in plaintext, maintaining complete confidentiality of user data throughout the copy detection process.
- Even cloud providers or system administrators cannot access the original video content.

2. Secure Computation in Encrypted Domain

- Enables matching and comparison operations directly on **encrypted features**, eliminating the need for decryption during analysis.
- Ensures **zero knowledge** processing, which significantly reduces the risk of data breaches.

3. Robustness to Multi-View Variability

- Incorporates **view-invariant feature extraction** techniques to accurately detect copies across different camera angles, lighting conditions, and occlusions.
- Increases detection accuracy in real-world scenarios such as surveillance or event recording from multiple viewpoints.

4. Cloud Compatibility and Scalability

- Designed for deployment in **cloud environments**, allowing secure storage and fast querying of large video datasets.
- Supports **horizontal scaling**, enabling the system to handle millions of video files and growing demands efficiently.

5. Reduced Risk of Unauthorized Access or Piracy

- As data is never decrypted during processing, it is inherently protected against **internal threats**, such as curious employees or compromised servers.
- Complies with data protection standards (e.g., **GDPR, HIPAA**) by design.

6. Low False Positive/Negative Rate

- Advanced feature extraction and encrypted-domain comparison reduce the chances of incorrect detection results, even when videos are transformed, edited, or partially altered.
- Supports detection of both **exact copies** and **content-preserving transformations**.

7. Automated and Scalable Detection Pipeline

- End-to-end automation from feature extraction to detection and alert generation ensures minimal manual intervention.
- Ideal for integration with large-scale content platforms (e.g., YouTube, Netflix, surveillance systems).

8. Future-Proof Cryptographic Design

- Homomorphic encryption is a foundational technology for **privacy-preserving AI** and **secure cloud computing**, making the system adaptable to future regulatory and technological requirements.

3.3 SYSTEM REQUIREMENTS

3.3.1 HARDWARE REQUIREMENTS

- System : Pentium IV 2.4 GHz.
- Hard Disk : 40 GB.
- Ram : 512 Mb.

3.3.2 SOFTWARE REQUIREMENTS

- Operating system : Windows.
- Coding Language : python

3.4 FUNCTIONAL REQUIREMENTS

1. **User Registration:** Allows new users to create an account.
2. **User Login:** Authenticates users to access the system.
3. **Video Upload:** Enables users to upload videos to the cloud.
4. **Feature Extraction:** Extracts important features from videos for comparison.
5. **Encrypted Comparison:** Compares videos in encrypted form to detect copies.
6. **Copy Detection Output:** Identifies and marks videos as unique or duplicate. preferences.

3.5 NON-FUNCTIONAL REQUIREMENTS

1. **PerformanceSecurity:** Ensures data is protected using encryption.
2. **Performance:** System should process videos efficiently with minimal delay.

3. **Scalability:** Can handle increasing number of videos and users.
4. **Reliability:** Provides consistent and accurate results.
5. **Usability:** Easy for users to interact with the system.
6. **Availability:** System should be accessible whenever needed.

3.5.1 ADVANTAGES

- Strong Privacy Protection
- Secure Cloud Usage
- Accurate Copy Detection
- Encrypted Domain Processing
- Scalability
- Real-time Detection
- Low Error Rate
- Piracy Prevention

3.5.2 DISADVANTAGES

- Accuracy High Computational Cost
- High Resource Requirement
- Complex Implementation
- Processing Delay
- Key Management Issues
- Storage Overhead
- Dependency on Feature Quality
- Limited Practical Deployment

3.5.3 KEY LEARNING

1. Understanding of **Homomorphic Encryption** for secure data processing
2. Knowledge of **video feature extraction techniques**
3. Experience with **cloud-based storage and processing**
4. Implementation of **privacy-preserving systems**
5. Integration of **machine learning and security concepts**
6. Hands-on experience with **Python and related libraries**

3.6 SYSTEM STUDY

3.6.1 FEASIBILITY STUDY:

The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential. Three key considerations involved in the feasibility analysis are:

- ECONOMICAL FEASIBILITY
- TECHNICAL FEASIBILITY
- OPERATIONAL FEASIBILITY

3.6.1.1 ECONOMICAL FEASIBILITY:

The proposed system is technically feasible as it uses well-established technologies like Python, machine learning, and homomorphic encryption. The required tools such as TensorFlow, NumPy, and cloud platforms are easily available and widely supported. The system can be developed using existing

frameworks and libraries, reducing development complexity. However, it requires good computational power for encryption and video processing, which can be handled using cloud resources.

3.6.1.2 TECHNICAL FEASIBILITY:

The project is economically feasible as it mainly relies on open-source technologies and tools, which reduces software costs. Cloud services provide scalable storage and processing at affordable prices, making it suitable for large datasets. There is no need for expensive proprietary software. Although there may be some cost for cloud usage and hardware, the overall development and maintenance cost is low compared to traditional systems.

3.6.1.3 OPERATIONAL FEASIBILITY:

The system is operationally feasible as it is designed with a simple and user-friendly interface. Users can easily register, upload videos, and detect duplicates without technical knowledge. The system automates most processes like feature extraction, encryption, and detection, reducing manual effort. It can be effectively used in real-world applications like video platforms and surveillance systems with minimal training.

3.6.2 STUDY OF THE SYSTEM

1. Numpy
2. Pandas
3. Matplotlib
4. Scikit-learn
5. Tensorflow

1. NumPy:

NumPy is a general-purpose array-processing package. It provides a high-performance multidimensional array object, and tools for working with these arrays.

It is the fundamental package for scientific computing with Python. It contains various features including these important ones:

- A powerful N-dimensional array object
- Sophisticated (broadcasting) functions
- Tools for integrating C/C++ and Fortran code
- Useful linear algebra, Fourier transform, and random number capabilities

Besides its obvious scientific uses, NumPy can also be used as an efficient multi-dimensional container of generic data. Arbitrary data-types can be defined using NumPy which allows NumPy to seamlessly and speedily integrate with a wide variety of databases.

2. Pandas:

Pandas is an open-source Python Library providing high-performance data manipulation and analysis tool using its powerful data structures. Python was majorly used for data munging and preparation. It had very little contribution towards data analysis. Pandas solved this problem. Using Pandas, we can accomplish five typical steps in the processing and analysis of data, regardless of the origin of data load, prepare, manipulate, model, and analyse. Python with Pandas is used in a wide range of fields including academic and commercial domains including finance, economics, Statistics, analytics, etc.

3. Matplotlib:

Matplotlib is a Python 2D plotting library which produces publication quality figures in a variety of hardcopy formats and interactive environments across platforms. Matplotlib can be used in Python scripts, the Python and IPython shells, the Jupyter notebook, web application servers, and four graphical user interface toolkits. Matplotlib tries to make easy things easy and hard things possible. You can generate plots, histograms, power spectra, bar charts, error charts, scatter plots, etc., with just a few lines of code. For examples, see the sample plots and thumbnail gallery.

For simple plotting the pyplot module provides a MATLAB-like interface, particularly when combined with IPython. For the power user, you have full control of line styles, font properties, axes properties, etc, via an object-oriented interface or via a set of functions familiar to MATLAB users.

4. Scikit –learn

Scikit-learn provides a range of supervised and unsupervised learning algorithms via a consistent interface in Python. It is licensed under a permissive simplified BSD license and is distributed under many Linux distributions, encouraging academic and commercial use. The library is built upon the SciPy (Scientific Python) that must be installed before you can use scikit-learn. This stack that includes:

- **NumPy**: Base n-dimensional array package
- **SciPy**: Fundamental library for scientific computing
- **Matplotlib**: Comprehensive 2D/3D plotting
- **IPython**: Enhanced interactive console
- **Sympy**: Symbolic mathematics
- **Pandas**: Data structures and analysis

Extensions or modules for SciPy are conventionally named SciKits. As such, the module provides learning algorithms and is named scikit-learn.

4. Tensor Flow

TensorFlow is a free and open-source software library for dataflow and differentiable programming across a range of tasks, including machine learning and deep learning. It was developed by Google Brain team and released under the Apache 2.0 open-source license in 2015. TensorFlow provides a flexible and efficient platform for building and deploying machine learning models, and can be used for a wide range of applications, including image recognition, natural language processing, and speech recognition. TensorFlow works by building computational graphs that represent the operations performed by the model, and then optimizing these graphs to run efficiently on a wide range of hardware, including CPUs, GPUs, and TPUs. TensorFlow is widely used in industry and academia, and has become one of the most popular machine learning libraries in the world.

3.7 MODULES

1. Data Collection Module
2. Data Preprocessing Module
3. Feature Extraction using Pre-trained Models
4. Ensemble Classification Module
5. Prediction and Result Display Module
6. Model Evaluation Module

MODULES DESCRIPTION

Data Collection Module:

This module is responsible for collecting video data from different sources. The collected data may include multi-view videos from cameras or cloud storage. It ensures that sufficient and relevant data is gathered for further processing.

Data Preprocessing Module:

In this module, the collected videos are cleaned and prepared for analysis. It includes tasks like resizing frames, removing noise, normalizing data, and converting videos into suitable formats. This step improves the quality of input data.

Feature Extraction using Pre- trained Models:

This module extracts important features from videos using pre-trained deep learning models. These features represent the content of videos in a compact form and help in identifying similarities between videos.

Ensemble Classification Module:

This module uses multiple machine learning models together to improve accuracy. By combining different classifiers, it provides better decision-making in identifying whether a video is duplicate or not.

Prediction and Result Display Module:

This module provides the final output of the system. It predicts whether the uploaded video is unique or duplicate and displays the result clearly to the user.

Model Evaluation Module:

This module evaluates the performance of the system using metrics like accuracy, precision, and recall. It helps in analysing how well the model is performing and improving it if needed.

4 SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE

Below diagram depicts the whole system architecture of Food calorie estimation and BMI prediction using Machine Learning and Deep Learning.

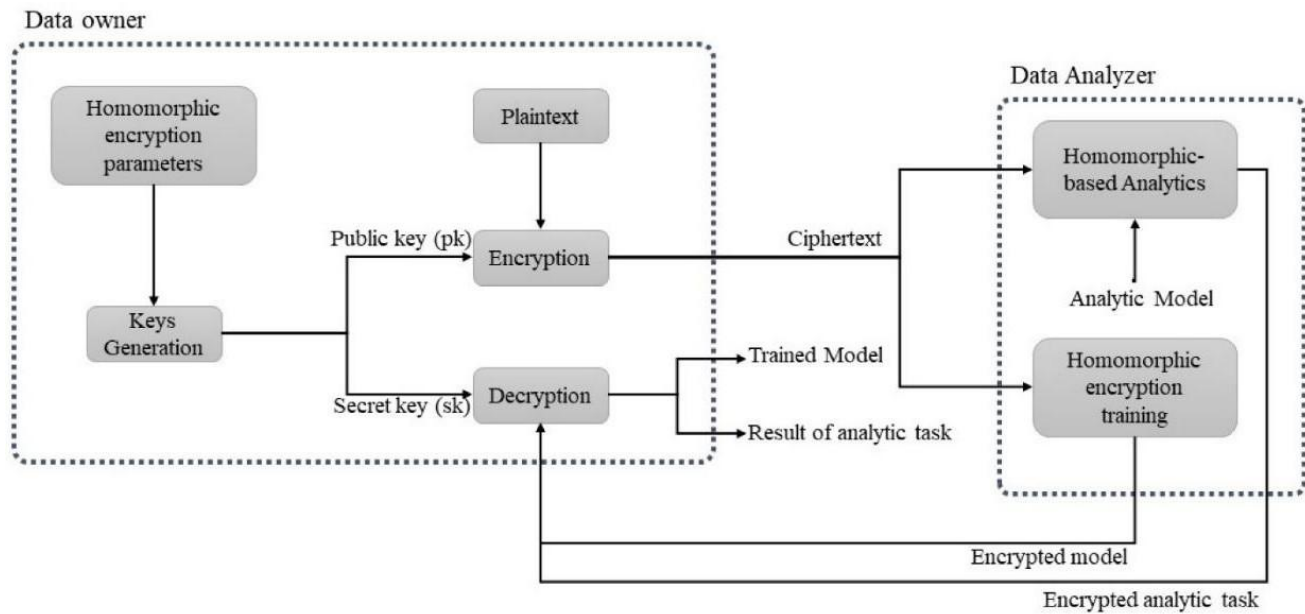


Fig. 4.1.1 System Architecture

4.2 UML REPRESENTATION

4.2.1 Use Case Diagram

- The use case diagram visualizes the conduct of a framework, subsystem, or class with the goal that clients can appreciate how to utilize that components thus that engineers can execute that component
- Below is the representation of the project, on which the system is developed.

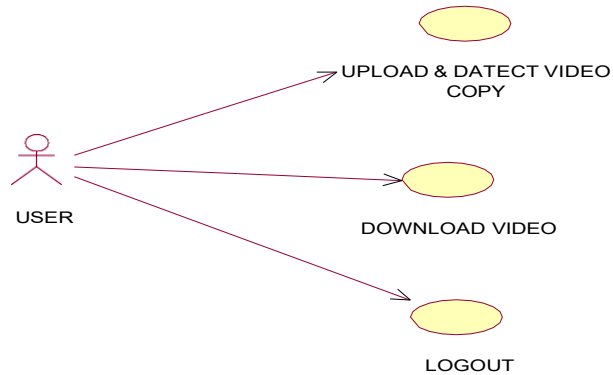


Fig. 4.2.1 Use Case Representation

4.2.2 Sequence Diagram

Sequence diagram is a construct of the Message Sequence Chart and conjointly it's a sort of interaction diagram that shows however processes operate with each other and in what o

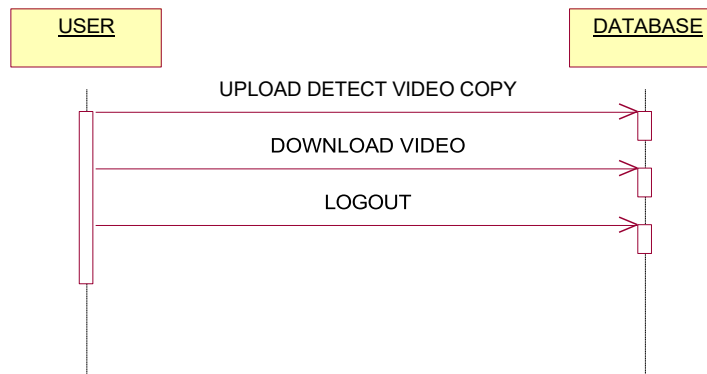


Fig. 4.2.2 Sequence Diagram

4.2.3 Activity Diagram

A graphical representations of work process of stepwise exercises and activities with support for decision, emphasis and simultaneousness, used to depict the business and operational well-ordered stream of parts in a framework furthermore demonstrates the general stream of control.

Activity Diagram: Video Workflow

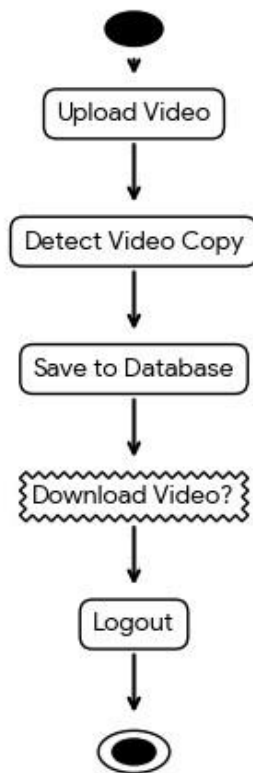


Fig. 4.2.3 Activity Diagram

4.1.1 Class Diagram

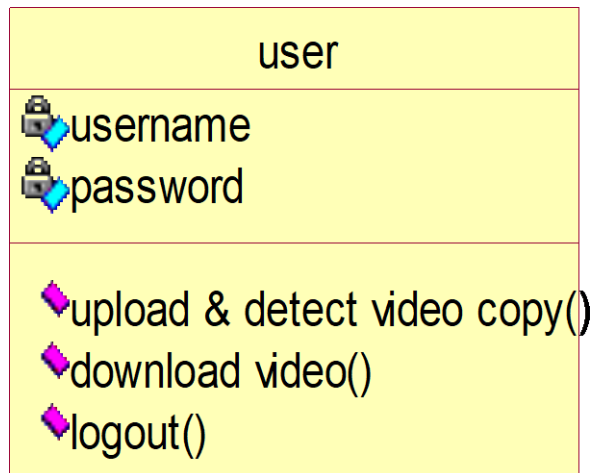


Fig. 4.2.4 Class Diagram

4.1.2 Component Diagram

Component Diagram: System Modules

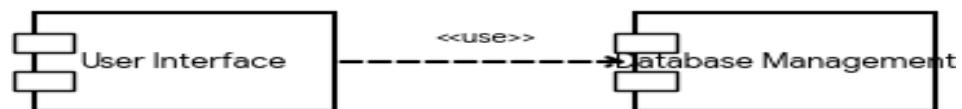


Fig. 4.2.5 Component Diagram

4.1.3 Collaboration Diagram

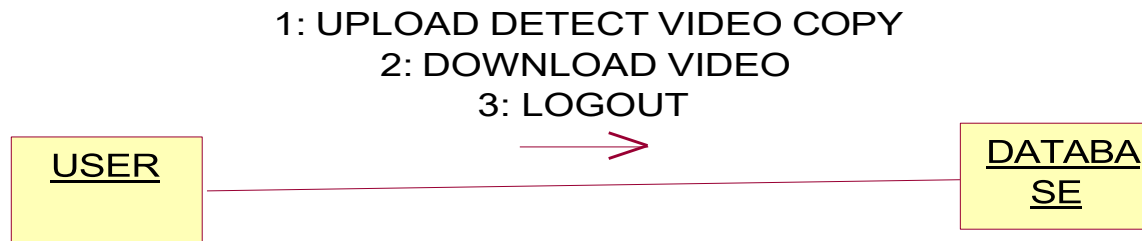


Fig. 4.2.6 Collaboration Diagram

4.1.4 Deployment Diagram

Deployment Diagram: Hardware Topology



Fig. 4.2.7 Deployment Diagram

4.3 INPUT AND OUTPUT DESIGN

INPUT DESIGN

The input design is the link between the information system and the user. It comprises the developing specification and procedures for data preparation and those steps are necessary to put transaction data in to a usable form for processing can be achieved by inspecting the computer to read data from a written or printed document or it can occur by having people keying the data directly into the system. The design of input focuses on controlling the amount of input required, controlling the errors, avoiding delay, avoiding extra steps and keeping the process simple. The input is designed in such a way so that it provides security and ease of use with retaining the privacy. Input Design considered the following things:

- What data should be given as input?
- How the data should be arranged or coded?
- The dialog to guide the operating personnel in providing input.
- Methods for preparing input validations and steps to follow when error occur.

OBJECTIVES

1. Input Design is the process of converting a user-oriented description of the input into a computer-based system. This design is important to avoid errors in the data input process and show the correct direction to the management for getting correct information from the computerized system.

2. It is achieved by creating user-friendly screens for the data entry to handle large volume of data. The goal of designing input is to make data entry easier and to be free from errors. The data entry screen is designed in such a way that all the data manipulates can be performed. It also provides record viewing facilities.

3. When the data is entered it will check for its validity. Data can be entered with the help of screens. Appropriate messages are provided as when needed so that the user will not be in maize of instant. Thus the objective of input design is to create an input layout that is easy to follow

OUTPUT DESIGN

A quality output is one, which meets the requirements of the end user and presents the information clearly. In any system results of processing are communicated to the users and to other system through outputs. In output design it is determined how the information is to be displaced for immediate need and also the hard copy output. It is the most important and direct source information to the user. Efficient and intelligent output design improves the system's relationship to help user decision-making.

1. Designing computer output should proceed in an organized, well thought out manner; the right output must be developed while ensuring that each output element is designed so that people will find the system can use easily and effectively. When analysis design computer output, they should Identify the specific output that is needed to meet the requirements.

2. Select methods for presenting information.

3. Create document, report, or other formats that contain information produced by the system.

The output form of an information system should accomplish one or more of the following objectives.

- Convey information about past activities, current status or projections of the
- Future.
- Signal important events, opportunities, problems, or warnings.
- Trigger an action.
- Confirm an action.

5. SYSTEM IMPLEMENTATION

5.1 SOURCE CODE

```
from django.contrib import admin

# Register your models here.

from django.apps import AppConfig

class VideocopyappConfig(AppConfig):
    name = 'VideoCopyApp'

from django.db import models

# Create your models here.

from django.test import TestCase

# Create your tests here.

from django.urls import path

from . import views

urlpatterns = [path("index.html", views.index, name="index"),
               path("UserLogin.html", views.UserLogin, name="UserLogin"),
               path("UserLoginAction", views.UserLoginAction, name="UserLoginAction"),
               path("Register.html", views.Register, name="Register"),
               path("RegisterAction", views.RegisterAction, name="RegisterAction"),
               path("UploadVideo.html", views.UploadVideo, name="UploadVideo"),
               path("UploadVideoAction", views.UploadVideoAction, name="UploadVideoAction"),
               path("DownloadVideo", views.DownloadVideo, name="DownloadVideo"),
               path("DownloadVideoAction", views.DownloadVideoAction, name="DownloadVideoAction"),

               ]

from django.shortcuts import render
from django.template import RequestContext
from django.contrib import messages
from django.http import HttpResponse
from django.conf import settings
import os
from django.core.files.storage import FileSystemStorage
import pymysql
from phe import paillier
import pickle
import os
from datetime import date

global username
```

```

def generateKeys():
    if os.path.exists('keys/fhe.pckl'):
        f = open('keys/fhe.pckl', 'rb')
        keys = pickle.load(f)
        f.close()
        public_key, private_key = keys
    else:
        # Generate a key pair (in a real application, you would likely reuse keys)
        public_key, private_key = phe.paillier.generate_paillier_keypair()
        keys = [public_key, private_key]
        f = open('keys/fhe.pckl', 'wb')
        pickle.dump(keys, f)
        f.close()
    return public_key, private_key

public_key, private_key = generateKeys()

def encrypt_video_chunk(chunk, public_key):
    chunk_int = int.from_bytes(chunk, byteorder='big')
    encrypted_chunk = public_key.encrypt(chunk_int)
    return encrypted_chunk

def readVideo(video_path, public_key):#function to read and encrypt video
    with open(video_path, "rb") as file:
        data = file.read() #read video
    file.close()
    data = data[0:300]
    chunks = encrypt_video_chunk(data, public_key)#encrypt and return video
    return chunks

def checkDuplicate(video_path):
    global public_key, private_key
    status = ""
    filename = ""
    chunks = readVideo(video_path, public_key)#reading FHE encrypted source video using public key as
chunks
    for root, dirs, directory in os.walk('VideoCopyApp/static/Videos'):
        for j in range(len(directory)):
            if status != "Duplicate":
                chunks1 = readVideo(root+"/"+directory[j], public_key) #getting database encrypted video as
chunks
                difference = chunks - chunks1 #now applying mathematical computation on homomorphic
encrypted video by calculating subtraction means difference
                decrypted_difference = private_key.decrypt(difference)
                if decrypted_difference > 0: #if difference > 0 video is unique
                    status = "Unique"
                elif decrypted_difference < 0: #if difference < 0 video is unique
                    status = "Unique"
                else: #if difference equals then video is duplicated
                    status = "Duplicate"
                    filename = directory[j]
                    break
            if status == "Duplicate":
                break
    if len(status) == 0:
        status = "Unique"

```

```

return status, filename

def DownloadVideoAction(request):
    if request.method == 'GET':
        name = request.GET.get('name', False)
        with open("VideoCopyApp/static/Videos/"+name, "rb") as file:
            data = file.read()
            file.close()
        response = HttpResponse(data,content_type='application/force-download')
        response['Content-Disposition'] = 'attachment; filename='+name
        return response

def DownloadVideo(request):
    if request.method == 'GET':
        global username
        output = ""
        output+="




```

```

myfile = request.FILES['t1'].read()
fname = request.FILES['t1'].name
if os.path.exists("VideoCopyApp/static/"+fname):
    os.remove("VideoCopyApp/static/"+fname)
with open("VideoCopyApp/static/"+fname, "wb") as file:
    file.write(myfile)
file.close()
status, filename = checkDuplicate("VideoCopyApp/static/"+fname)
output = ""
dd = str(date.today())
if status == "Unique":
    os.remove("VideoCopyApp/static/"+fname)
    with open("VideoCopyApp/static/Videos/"+fname, "wb") as file:
        file.write(myfile)
    file.close()
    db_connection = pymysql.connect(host='127.0.0.1',port = 3306,user = 'root', password = 'root',
database = 'videocopy',charset='utf8')
    db_cursor = db_connection.cursor()
    student_sql_query = "INSERT INTO videos
VALUES('"+username+"','"+fname+"','"+dd+"','"+status+"','"+fname+"')"
    db_cursor.execute(student_sql_query)
    db_connection.commit()
    output = "<font size=3 color=blue>Uploaded video "+fname+" detected as unique</font>"
else:
    os.remove("VideoCopyApp/static/"+fname)
    db_connection = pymysql.connect(host='127.0.0.1',port = 3306,user = 'root', password = 'root',
database = 'videocopy',charset='utf8')
    db_cursor = db_connection.cursor()
    student_sql_query = "INSERT INTO videos
VALUES('"+username+"','"+fname+"','"+dd+"','"+status+"','"+filename+"')"
    db_cursor.execute(student_sql_query)
    db_connection.commit()
    output = "<font size=3 color=red>Uploaded video = "+fname+" found duplicate with existing cloud
video = "+filename+"</font>"
context= {'data':output}
return render(request, 'UserScreen.html', context)

def RegisterAction(request):
    if request.method == 'POST':
        global username
        username = request.POST.get('t1', False)
        password = request.POST.get('t2', False)
        contact = request.POST.get('t3', False)
        email = request.POST.get('t4', False)
        address = request.POST.get('t5', False)
        output = "none"
        con = pymysql.connect(host='127.0.0.1',port = 3306,user = 'root', password = 'root', database =
'veideocopy',charset='utf8')
        with con:
            cur = con.cursor()
            cur.execute("select username FROM register")
            rows = cur.fetchall()
            for row in rows:
                if row[0] == username:
                    output = username+" Username already exists"
                    break

```

```

        if output == "none":
            db_connection = pymysql.connect(host='127.0.0.1',port = 3306,user = 'root', password = 'root',
            database = 'videocopy',charset='utf8')
            db_cursor = db_connection.cursor()
            student_sql_query = "INSERT INTO register
VALUES('"+username+"','"+password+"','"+contact+"','"+email+"','"+address+"')"
            db_cursor.execute(student_sql_query)
            db_connection.commit()
            print(db_cursor.rowcount, "Record Inserted")
            if db_cursor.rowcount == 1:
                output = "Signup process completed. Login to perform video copy detection activities"
            context= {'data':output}
            return render(request, 'Register.html', context)

```

```

def UserLoginAction(request):
    global username
    if request.method == 'POST':
        global username
        status = "none"
        users = request.POST.get('t1', False)
        password = request.POST.get('t2', False)
        con = pymysql.connect(host='127.0.0.1',port = 3306,user = 'root', password = 'root', database =
        'videocopy',charset='utf8')
        with con:
            cur = con.cursor()
            cur.execute("select username,password FROM register")
            rows = cur.fetchall()
            for row in rows:
                if row[0] == users and row[1] == password:
                    username = users
                    status = "success"
                    break
            if status == 'success':
                context= {'data': 'Welcome '+username}
                return render(request, "UserScreen.html", context)
            else:
                context= {'data': 'Invalid username'}
                return render(request, 'UserLogin.html', context)

```

```

def Register(request):
    if request.method == 'GET':
        return render(request, 'Register.html', {})

```

```

def UserLogin(request):
    if request.method == 'GET':
        return render(request, 'UserLogin.html', {})

```

```

def index(request):
    if request.method == 'GET':
        return render(request, 'index.html', {})

```

```

{% load static %}
<html>
<head>

```



```

<meta http-equiv="content-type" content="text/html; charset=utf-8" />
<title>Video Copy Detection</title>
<meta name="keywords" content="" />
<meta name="Premium Series" content="" />
<link href="{% static 'default.css' %}" rel="stylesheet" type="text/css" media="screen" />
<style>
body {
  background-image: url("{% static 'images/back.jpg' %}");
}
</style>
</head>
<body>
<!-- start header -->
<div id="header">
  <div id="logo">
    <h1><span><center><font size="4" color="white"><br/>Homomorphic Encryption Based Video
Copy Detection in Multi View Videos Stored in a Cloud</font>

</center></span></h1>

    </div>
    <div id="menu">
      <ul id="main">
        <li class="current_page_item"><a href="{% url 'index' %}"><font size="4"
color="white">Home</a></li><td></td>
        <li><a href="{% url 'UserLogin' %}"><font size="4" color="white">User Login
Here</a></li><td></td>
        <li><a href="{% url 'Register' %}"><font size="4" color="white">New User Signup
Here</a></li><td></td>
        <td></td>

      </ul>

    </div>

  </div>
</div>
<div id="wrapper">
  <!-- start page -->
  <div id="page">

    <!-- start content -->
    <div id="content">
      <div class="flower"></center></div>
      <br/><br/>
      <center><font size="3" color="black">Homomorphic Encryption Based Video Copy Detection in
Multi View Videos Stored in a Cloud</font></center>
      <center>

        <table align="center">
          <tr>
            <h3>
          </p></h3>
        </table>

```

```

</center><br /><div class="post">

    <div class="entry">

        </div>
    </div>
</div>
<!-- end content -->
<!-- start sidebars -->

<!-- end sidebars -->
<div style="clear: both;">&nbsp;</div>
</div>
<!-- end page -->
</div>
</body>
</html>

```

```

{% load static %}
<html>
<head>
<meta http-equiv="content-type" content="text/html; charset=utf-8" />
<title>Video Copy Detection</title>
<meta name="keywords" content="" />
<meta name="Premium Series" content="" />
<link href="{% static 'default.css' %}" rel="stylesheet" type="text/css" media="screen" />
<script language="javascript">
    function validate(formObj)
    {
        if(formObj.t1.value.length==0)
        {
            alert("Please Enter Uername");
            formObj.t1.focus();
            return false;
        }
        if(formObj.t2.value.length==0)
        {
            alert("Please Enter Password");
            formObj.t2.focus();
            return false;
        }
        if(formObj.t3.value.length==0)
        {
            alert("Please Enter Contact No");
            formObj.t3.focus();
            return false;
        }
        if(formObj.t3.value.length==0)
        {
            alert("Please Enter Contact No");
            formObj.t3.focus();
            return false;
        }
        if(isNaN(formObj.t3.value)){

```

```

        alert("Contact No must be numeric");
        formObj.t3.focus();
        return false;
    }
    if(formObj.t4.value.length==0)
    {
        alert("Please Enter Email Id");
        formObj.t4.focus();
        return false;
    }
    var filter = /^[a-zA-Z0-9_\.\\-]+\@([a-z]+\.)+(com)+$/;
    if (!filter.test(formObj.t4.value)) {
        window.alert('Please enter valid email address');
        formObj.t4.focus();
        return false;
    }
    if(formObj.t5.value.length==0)
    {
        alert("Please Enter address");
        formObj.t5.focus();
        return false;
    }
    return true;
}
</script>
<style>
body {
    background-image: url("{% static 'images/back.jpg' %}");
}
</style>
</head>
<body>
<!-- start header -->
<div id="header">
    <div id="logo">
        <h1><span><center><font size="5" color="white"><br/>Homomorphic Encryption Based Video
Copy Detection in Multi View Videos Stored in a Cloud</font>
</center></span></h1>

    </div>
    <div id="menu">
        <ul id="main">
            <li class="current_page_item"><a href="{% url 'index' %}"><font size="4"
color="white">Home</a></li><td></td>
            <li><a href="{% url 'UserLogin' %}"><font size="4" color="white">User Login
Here</a></li><td></td>
            <li><a href="{% url 'Register' %}"><font size="4" color="white">New User Signup
Here</a></li><td></td>
            <td></td>

        </ul>

    </div>

</div>

</div>

```

```

<div id="wrapper">
  <!-- start page -->
  <div id="page">
    <form name="f1" method="post" action="{% url 'RegisterAction' %}" onsubmit="return
validate(this);">
      {% csrf_token %}
      <!-- start content -->
      <div id="content">
        <div class="flower"></center></div>
        <br/><br/>
        <div class="flower"><center><h1>
        <font size="" align="center">
        New User Signup Page</font><h1></div>

        <center><font face="verdana" color="blue">{{ data|safe }}</center></font>
      <br/>
      <table align="center">
        <tr><td><font size="3" color="black">Username</td><td><input type="text" name="t1"
size="30"/></td></tr>
        <tr><td><font size="3" color="black">Password</td><td><input type="password" name="t2"
size="30"/></td></tr>
        <tr><td><font size="3" color="black">Contact&nbsp;No</td><td><input type="text" name="t3"
size="15"/></td></tr>
        <tr><td><font size="3" color="black">Email&nbsp;ID</td><td><input type="text" name="t4"
size="30"/></td></tr>
        <tr><td><font size="3" color="black">Address</td><td><input type="text" name="t5"
size="40"/></td></tr>
        </td></tr>

      </td></tr>
      <tr><td></td><td><input type="submit" value="Submit"></td>
      </td></tr></table>
    </form>
    </table>
    </center><br /><div class="post">

      <div class="entry">

        </div>
      </div>
    </div>
    <!-- end content -->
    <!-- start sidebars -->

    <!-- end sidebars -->
    <div style="clear: both;">&nbsp;</div>
  </div>
  <!-- end page -->
</div>
</body>
</html>

```

```

{% load static %}
<html>
<head>
<meta http-equiv="content-type" content="text/html; charset=utf-8" />
<title>Video Copy Detection</title>
<meta name="keywords" content="" />
<meta name="Premium Series" content="" />
<link href="{% static 'default.css' %}" rel="stylesheet" type="text/css" media="screen" />
<script language="javascript">

    function validate(formObj)
    {
        if(formObj.t1.value.length==0)
        {
            alert("Please upload test data file");
            formObj.t1.focus();
            return false;
        }

        return true;
    }
</script>
<style>
body {
    background-image: url("{% static 'images/back.jpg' %}");
}
</style>
</head>
<body>
<!-- start header -->
<div id="header">
    <div id="logo">
        <h1><span><center><font size="5" color="white"><br/>Homomorphic Encryption Based Video
Copy Detection in Multi View Videos Stored in a Cloud</font>
</center></span></h1>

    </div>
    <div id="menu">
        <ul id="main">
            <li class="current_page_item"><a href="{% url 'UploadVideo' %}"><font size="3"
color="white">Upload & Detect Video Copy</a></li><td></td>
            <li><a href="{% url 'DownloadVideo' %}"><font size="4" color="white">Download
Video</a></li><td></td>
            <li><a href="{% url 'index' %}"><font size="3" color="white">Logout</a></li><td></td>
            <td></td>

        </ul>

    </div>

</div>

</div>

<div id="wrapper">
    <!-- start page -->

```



```

    if(formObj.t1.value.length==0)
    {
        alert("Please Enter username");
        formObj.t1.focus();
        return false;
    }
    if(formObj.t2.value.length==0)
    {
        alert("Please Enter password");
        formObj.t2.focus();
        return false;
    }
    formObj.actionUpdateData.value="update";
    return true;
}
</script>
<style>
body {
    background-image: url("{% static 'images/back.jpg' %}");
}
</style>
</head>
<body>
<!-- start header -->
<div id="header">
    <div id="logo">
        <h1><span><center><font size="5" color="white"><br/>Homomorphic Encryption Based Video
Copy Detection in Multi View Videos Stored in a Cloud</font>
</center></span></h1>

    </div>
    <div id="menu">
        <ul id="main">
            <li class="current_page_item"><a href="{% url 'index' %}"><font size="4"
color="white">Home</a></li><td></td>
            <li><a href="{% url 'UserLogin' %}"><font size="4" color="white">User Login
Here</a></li><td></td>
            <li><a href="{% url 'Register' %}"><font size="4" color="white">New User Signup
Here</a></li><td></td>
            <td></td>

        </ul>

    </div>
</div>

<div id="wrapper">
    <!-- start page -->
    <div id="page">
        <form name="f1" method="post" action="{% url 'UserLoginAction' %}" OnSubmit="return
validate(this)">
            {% csrf_token %}
            <!-- start content -->
            <div id="content">

```

```

        <div class="flower"></center></div>
        <br/><br/>
        <div class="flower"><center><h1>
        <font size="" align="center">
        User Login Page</font><h1></div>

        <center><font face="verdana" color="blue">{{ data|safe }}</center></font>
    <br/>
        <table align="center">
    <tr><td><font size="3" color="black">Username</td><td><input type="text" name="t1"
    size="30"/></td></tr>
    <tr><td><font size="3" color="black">Password</td><td><input type="password" name="t2"
    size="30"/></td></tr>

</td></tr>

</td></tr>
    <tr><td></td><td><input type="submit" value="Login"></td>
</td></tr></table>
</form>
    </table>
    </center><br /><div class="post">

        <div class="entry">

            </div>
        </div>
    </div>
    <!-- end content -->
    <!-- start sidebars -->

    <!-- end sidebars -->
    <div style="clear: both;">&nbsp;</div>
</div>
<!-- end page -->
</div>
</body>
</html>

```

```

{% load static %}
<html>
<head>
<meta http-equiv="content-type" content="text/html; charset=utf-8" />
<title>Video Copy Detection</title>
<meta name="keywords" content="" />
<meta name="Premium Series" content="" />
<link href="{% static 'default.css' %}" rel="stylesheet" type="text/css" media="screen" />
<style>
body {
    background-image: url("{% static 'images/back.jpg' %}");
}
</style>
</head>

```



```

<body>
<!-- start header -->
<div id="header">
  <div id="logo">
    <h1><span><center><font size="5" color="white"><br/>Homomorphic Encryption Based Video
Copy Detection in Multi View Videos Stored in a Cloud</font>

</center></span></h1>

  </div>
  <div id="menu">
    <ul id="main">
      <li class="current_page_item"><a href="{% url 'UploadVideo' %}"><font size="3"
color="white">Upload & Detect Video Copy</a></li><td></td>
      <li><a href="{% url 'DownloadVideo' %}"><font size="4" color="white">Download
Video</a></li><td></td>
      <li><a href="{% url 'index' %}"><font size="3" color="white">Logout</a></li><td></td>
      <td></td>

    </ul>

  </div>

</div>
<div id="wrapper">
  <!-- start page -->
  <div id="page">

    <!-- start content -->
    <div id="content">
      <div class="flower"></center></div>
      <br/><br/>

      <center><font face="verdana" color="black">{{ data|safe }}</center></font><br/>
      

      <center>

        <table align="center">
          <tr>
            <h3>
          </p></h3>
        </table>
      </center><br /><div class="post">

        <div class="entry">

          </div>
        </div>
      </div>
      <!-- end content -->
      <!-- start sidebars -->

      <!-- end sidebars -->
      <div style="clear: both;">&nbsp;</div>

```

```

</div>
<!-- end page -->
</div>
</body>
</html>

```

```

#!/usr/bin/env python
import os
import sys

if __name__ == '__main__':
    os.environ.setdefault('DJANGO_SETTINGS_MODULE', 'VideoCopy.settings')
    try:
        from django.core.management import execute_from_command_line
    except ImportError as exc:
        raise ImportError(
            "Couldn't import Django. Are you sure it's installed and "
            "available on your PYTHONPATH environment variable? Did you "
            "forget to activate a virtual environment?"
        ) from exc
    execute_from_command_line(sys.argv)

```

```

pip install numpy==1.21.6
pip install Django==2.1.7
pip install PyMySQL==0.9.3
pip install phe==1.5.0

```

```

python manage.py runserver
pause

```

```

from phe import paillier
import pickle
import os

```

```

def generateKeys():
    if os.path.exists('keys/fhe.pckl'):
        f = open('keys/fhe.pckl', 'rb')
        keys = pickle.load(f)
        f.close()
        public_key, private_key = keys
    else:
        # Generate a key pair (in a real application, you would likely reuse keys)
        public_key, private_key = phe.paillier.generate_paillier_keypair()
        keys = [public_key, private_key]
        f = open('keys/fhe.pckl', 'wb')
        pickle.dump(keys, f)
        f.close()
    return public_key, private_key

```

```

def encrypt_video_chunk(chunk, public_key):

```

```

chunk_int = int.from_bytes(chunk, byteorder='big')
encrypted_chunk = public_key.encrypt(chunk_int)
return encrypted_chunk

def readVideo(video_path, public_key):
    with open(video_path, "rb") as file:
        data = file.read()
    file.close()
    data = data[0:300]
    chunks = encrypt_video_chunk(data, public_key)
    return chunks

public_key, private_key = generateKeys()
chunks = readVideo("testVideos/11.mp4", public_key)
for root, dirs, directory in os.walk('Videos'):
    for j in range(len(directory)):
        chunks1 = readVideo(root+"/"+directory[j], public_key)
        difference = chunks - chunks1
        decrypted_difference = private_key.decrypt(difference)
        if decrypted_difference > 0:
            print("Video is unique "+directory[j])
        elif decrypted_difference < 0:
            print("Video is unique "+directory[j])
        else:
            print("Video is duplicate "+directory[j])

```

5.2 ALGORITHMS

1. Homomorphic Encryption

This is the core algorithmic component of the system. It allows the cloud server to perform operations directly on encrypted video data without needing to decrypt it.

- **Similarity Comparisons:** The system uses mathematical properties of homomorphic encryption to perform similarity checking and feature comparisons securely in an untrusted cloud environment.
- **Confidentiality:** It ensures end-to-end data confidentiality because the raw video content is never accessed by the cloud.

2. Feature Extraction & Matching

The system employs specific algorithms to represent and compare video content:

- **Robust Feature Extraction:** Videos are preprocessed on the user side to extract robust features before they are encrypted and uploaded.

- **Spatial and Temporal Analysis:** Similar to existing systems, this project analyzes both spatial and temporal features to identify duplicates or near-duplicates.

3. Multi-View Feature Alignment

Because the project deals with multi-view videos (captured from multiple angles), it uses alignment techniques to integrate these perspectives:

- **Perspective Integration:** This technique integrates information from various camera angles into integrated feature models to enhance detection accuracy.

4. System Implementation Details

- **Coding Language:** The algorithms are implemented using **Python**.
- **Similarity Search:** The cloud utilizes similarity comparisons on the encrypted features to identify copied content across different views.

5.3 SOFTWARE ENVIRONMENT

5.3.1 PYTHON

Python is a high-level, interpreted, interactive and object-oriented scripting language. Python is designed to be highly readable. It uses English keywords frequently where as other languages use punctuation, and it has fewer syntactical constructions than other languages.

Python is Interpreted – Python is processed at runtime by the interpreter. You do not need to compile your program before executing it. This is similar to PERL and PHP.

Python is Interactive – You can actually sit at a Python prompt and interact with the interpreter directly to write your programs.

Python is Object-Oriented – Python supports Object-Oriented style or technique of programming that encapsulates code within objects.

Python is a Beginner's Language – Python is a great language for the beginner-level programmers and supports the development of a wide range of applications from simple text processing to WWW browsers to games.

Advantages of Python :-

Let's see how Python dominates over other languages.

1. Extensive Libraries

Python downloads with an extensive library and it *contain code for various purposes like regular expressions, documentation-generation, unit-testing, web browsers, threading, databases, CGI, email, image manipulation, and more*. So, we don't have to write the complete code for that manually.

2. Extensible

As we have seen earlier, Python can be **extended to other languages**. You can write some of your code in languages like C++ or C. This comes in handy, especially in projects.

3. Embeddable

Complimentary to extensibility, Python is embeddable as well. You can put your Python code in your source code of a different language, like C++. This lets us add **scripting capabilities** to our code in the other language.

4. Improved Productivity

The language's simplicity and extensive libraries render programmers **more productive** than languages like Java and C++ do. Also, the fact that you need to write less and get more things done.

5. IOT Opportunities

Since Python forms the basis of new platforms like Raspberry Pi, it finds the future bright for the Internet Of Things. This is a way to connect the language with the real world.

6. Simple and Easy

When working with Java, you may have to create a class to print '**Hello World**'. But in Python, just a print statement will do. It is also quite **easy to learn, understand, and code**. This is why when people pick up Python, they have a hard time adjusting to other more verbose languages like Java.

7. Readable

Because it is not such a verbose language, reading Python is much like reading English. This is the reason why it is so easy to learn, understand, and code. It also does not need curly braces to define blocks, and **indentation is mandatory**. This further aids the readability of the code.

8. Object-Oriented

This language supports both the **procedural and object-oriented** programming paradigms. While functions help us with code reusability, classes and objects let us model the real world. A class allows the **encapsulation of data** and functions into one.

9. Free and Open-Source

Like we said earlier, Python is **freely available**. But not only can you **download Python** for free, but you can also download its source code, make changes to it, and even distribute it. It downloads with an extensive collection of libraries to help you with your tasks.

10. Portable

When you code your project in a language like C++, you may need to make some changes to it if you want to run it on another platform. But it isn't the same with Python. Here, you need to **code only once**, and you can run it anywhere. This is called **Write Once Run Anywhere (WORA)**. However, you need to be careful enough not to include any system-dependent features.

11. Interpreted

Lastly, we will say that it is an interpreted language. Since statements are executed one by one, **debugging is easier** than in compiled languages.

Any doubts till now in the advantages of Python? Mention in the comment section.

Disadvantages of Python

So far, we've seen why Python is a great choice for your project. But if you choose it, you should be aware of its consequences as well. Let's now see the downsides of choosing Python over another language.

1. Speed Limitations

We have seen that Python code is executed line by line. But since Python is interpreted, it often results in **slow execution**. This, however, isn't a problem unless speed is a focal point for the project. In other words, unless high speed is a requirement, the benefits offered by Python are enough to distract us from its speed limitations.

2. Weak in Mobile Computing and Browsers

While it serves as an excellent server-side language, Python is much rarely seen on the **client-side**. Besides that, it is rarely ever used to implement smartphone-based applications. One such application is called **Carbannelle**.

The reason it is not so famous despite the existence of Brython is that it isn't that secure.

3. Design Restrictions

As you know, Python is **dynamically-typed**. This means that you don't need to declare the type of variable while writing the code. It uses **duck-typing**. But wait, what's that? Well, it just means that if it looks like a duck, it must be a duck. While this is easy on the programmers during coding, it can **raise run-time errors**.

4. Underdeveloped Database Access Layers

Compared to more widely used technologies like **JDBC (Java DataBase Connectivity)** and **ODBC (Open DataBase Connectivity)**, Python's database access layers are a bit underdeveloped. Consequently, it is less often applied in huge enterprises.

5. Simple

No, we're not kidding. Python's simplicity can indeed be a problem. Take my example. I don't do Java, I'm more of a Python person. To me, its syntax is so simple that the verbosity of Java code seems unnecessary.

This was all about the Advantages and Disadvantages of Python Programming Language

5.3.2 Machine Learning

What is Machine Learning :-

Before we take a look at the details of various machine learning methods, let's start by looking at what machine learning is, and what it isn't. Machine learning is often categorized as a subfield of artificial intelligence, but I find that categorization can often be misleading at first brush. The study of machine learning certainly arose from research in this context, but in the data science application of machine learning methods, it's more helpful to think of machine learning as a means of *building models of data*.

Fundamentally, machine learning involves building mathematical models to help understand data. "Learning" enters the fray when we give these models *tunable parameters* that can be adapted to observed data; in this way the program can be considered to be "learning" from the data.

Once these models have been fit to previously seen data, they can be used to predict and understand aspects of newly observed data. I'll leave to the reader the more philosophical digression regarding the extent to which this type of mathematical, model-based "learning" is similar to the "learning" exhibited by the human brain. Understanding the problem setting in machine learning is essential to using these tools effectively, and so we will start with some broad categorizations of the types of approaches we'll discuss here.

Categories Of Machine Learning :-

At the most fundamental level, machine learning can be categorized into two main types: supervised learning and unsupervised learning.

Supervised learning involves somehow modeling the relationship between measured features of data and some label associated with the data; once this model is determined, it can be used to apply labels to new, unknown data. This is further subdivided into *classification* tasks and *regression* tasks: in classification, the labels are discrete categories, while in regression, the labels are continuous quantities. We will see examples of both types of supervised learning in the following section.

Unsupervised learning involves modelling the features of a dataset without reference to any label, and is often described as "letting the dataset speak for itself." These models include tasks such as *clustering* and *dimensionality reduction*.

Clustering algorithms identify distinct groups of data, while dimensionality reduction algorithms search for more succinct representations of the data. We will see examples of both types of unsupervised learning in the following section.

Need for Machine Learning

Human beings, at this moment, are the most intelligent and advanced species on earth because they can think, evaluate and solve complex problems. On the other side, AI is still in its initial stage and haven't surpassed human intelligence in many aspects. Then the question is that what is the need to make machine learn? The most suitable reason for doing this is, "to make decisions, based on data, with efficiency and scale".

Lately, organizations are investing heavily in newer technologies like Artificial Intelligence, Machine Learning and Deep Learning to get the key information from data to perform several real-world tasks and solve problems. We can call it data-driven decisions taken by machines, particularly to automate the process. These data-driven decisions can be used, instead of using programming logic, in the problems that cannot be programmed inherently. The fact is that we can't do without human intelligence, but other aspect is that we all need to solve real-world problems with efficiency at a huge scale. That is why the need for machine learning arises.

Challenges in Machines Learning :-

While Machine Learning is rapidly evolving, making significant strides with cybersecurity and autonomous cars, this segment of AI as whole still has a long way to go. The reason behind is that ML has not been able to overcome number of challenges. The challenges that ML is facing currently are –

Quality of data – Having good-quality data for ML algorithms is one of the biggest challenges. Use of low-quality data leads to the problems related to data preprocessing and feature extraction.

Time-Consuming task – Another challenge faced by ML models is the consumption of time especially for data acquisition, feature extraction and retrieval.

Lack of specialist persons – As ML technology is still in its infancy stage, availability of expert resources is a tough job.

No clear objective for formulating business problems – Having no clear objective and well-defined goal for business problems is another key challenge for ML because this technology is not that mature yet.

Issue of overfitting & underfitting – If the model is overfitting or underfitting, it cannot be represented well for the problem.

Curse of dimensionality – Another challenge ML model faces is too many features of data points. This can be a real hindrance.

Difficulty in deployment – Complexity of the ML model makes it quite difficult to be deployed in real life.

Applications of Machines Learning :-

Machine Learning is the most rapidly growing technology and according to researchers we are in the golden year of AI and ML. It is used to solve many real-world complex problems which cannot be solved with traditional approach. Following are some real-world applications of ML –

- Emotion analysis
- Sentiment analysis
- Error detection and prevention
- Weather forecasting and prediction
- Stock market analysis and forecasting
- Speech synthesis
- Speech recognition
- Customer segmentation
- Object recognition
- Fraud detection

- Fraud prevention
- Recommendation of products to customer in online shopping

Advantages of Machine learning :-

1. Easily identifies trends and patterns -

Machine Learning can review large volumes of data and discover specific trends and patterns that would not be apparent to humans. For instance, for an e-commerce website like Amazon, it serves to understand the browsing behaviors and purchase histories of its users to help cater to the right products, deals, and reminders relevant to them. It uses the results to reveal relevant advertisements to them.

2. No human intervention needed (automation)

With ML, you don't need to babysit your project every step of the way. Since it means giving machines the ability to learn, it lets them make predictions and also improve the algorithms on their own. A common example of this is anti-virus softwares; they learn to filter new threats as they are recognized. ML is also good at recognizing spam.

3. Continuous Improvement

As **ML algorithms** gain experience, they keep improving in accuracy and efficiency. This lets them make better decisions. Say you need to make a weather forecast model. As the amount of data you have keeps growing, your algorithms learn to make more accurate predictions faster.

4. Handling multi-dimensional and multi-variety data

Machine Learning algorithms are good at handling data that are multi-dimensional and multi-variety, and they can do this in dynamic or uncertain environments.

5. Wide Applications

You could be an e-tailer or a healthcare provider and make ML work for you. Where it does apply, it holds the capability to help deliver a much more personal experience to customers while also targeting the right customers.

Disadvantages of Machine Learning :-

1. Data Acquisition

Machine Learning requires massive data sets to train on, and these should be inclusive/unbiased, and of good quality. There can also be times where they must wait for new data to be generated.

2. Time and Resources

ML needs enough time to let the algorithms learn and develop enough to fulfill their purpose with a considerable amount of accuracy and relevancy. It also needs massive resources to function. This can mean additional requirements of computer power for you.

3. Interpretation of Results

Another major challenge is the ability to accurately interpret results generated by the algorithms. You must also carefully choose the algorithms for your purpose.

4. High error-susceptibility

Machine Learning is autonomous but highly susceptible to errors. Suppose you train an algorithm with data sets small enough to not be inclusive. You end up with biased predictions coming from a biased training set. This leads to irrelevant advertisements being displayed to customers. In the case of ML, such blunders can set off a chain of errors that can go undetected for long periods of time. And when they do get noticed, it takes quite some time to recognize the source of the issue, and even longer to correct it.

Python Development Steps :-

Guido Van Rossum published the first version of Python code (version 0.9.0) at alt.sources in February 1991. This release included already exception handling, functions, and the core data types of list, dict, str and others. It was also object oriented and had a module system. Python version 1.0 was released in January 1994. The major new features included in this release were the functional programming tools lambda, map, filter and reduce, which Guido Van Rossum never liked. Six and a half years later in October 2000, Python 2.0 was introduced. This release included list comprehensions, a full garbage collector and it was supporting unicode. Python flourished for another 8 years in the versions 2.x before the next major release as Python 3.0 (also known as "Python 3000" and "Py3K") was released. Python 3 is not backwards compatible with

Python 2.x.

The emphasis in Python 3 had been on the removal of duplicate programming constructs and modules, thus fulfilling or coming close to fulfilling the 13th law of the Zen of Python: "There should be one -- and preferably only one -- obvious way to do it." Some changes in Python 7.3:

- Print is now a function
- Views and iterators instead of lists
- The rules for ordering comparisons have been simplified. E.g. a heterogeneous list cannot be sorted, because all the elements of a list must be comparable to each other.
- There is only one integer type left, i.e. int. long is int as well.
- The division of two integers returns a float instead of an integer. "/" can be used to have the "old" behaviour.
- Text Vs. Data Instead Of Unicode Vs. 8-bit

Python

Python is an interpreted high-level programming language for general-purpose programming. Created by Guido van Rossum and first released in 1991, Python has a design philosophy that emphasizes code readability, notably using significant whitespace.

Python features a dynamic type system and automatic memory management. It supports multiple programming paradigms, including object-oriented, imperative, functional and procedural, and has a large and comprehensive standard library.

- Python is Interpreted – Python is processed at runtime by the interpreter. You do not need to compile your program before executing it. This is similar to PERL and PHP.
- Python is Interactive – you can actually sit at a Python prompt and interact with the interpreter directly to write your programs.

Python also acknowledges that speed of development is important. Readable and terse code is part of this, and so is access to powerful constructs that avoid tedious repetition of code. Maintainability also ties into this may be an all but useless metric, but it does say something about how much code you have to scan, read and/or understand to troubleshoot problems or tweak behaviors. This speed of development, the ease with which a programmer of other languages can pick up basic Python skills and the huge standard library is key to another area where Python excels. All its tools have been quick to implement, saved a lot of time, and several of them have later been patched and updated by people with no Python background - without Version control breaking.

6. TESTING

SYSTEM TEST

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

6.1 TYPES OF TESTS

6.1.1 Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application. It is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

6.1.2 Functional test

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

- Valid Input : identified classes of valid input must be accepted.
- Invalid Input : identified classes of invalid input must be rejected.

- Functions : identified functions must be exercised.
- Output : identified classes of application outputs must be exercised.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

6.1.3 System Test

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

6.1.4 White Box Testing

White Box Testing is a testing in which in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is purpose. It is used to test areas that cannot be reached from a black box level.

6.1.5 Black Box Testing

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box .you cannot “see” into it. The test provides inputs and responds to outputs without considering how the software works.

Unit Testing

Unit testing is usually conducted as part of a combined code and unit test phase of the software lifecycle, although it is not uncommon for coding and unit testing to be conducted as two distinct phases.

Test strategy and approach

Field testing will be performed manually and functional tests will be written in detail.

Test objectives

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

Features to be tested

- Verify that the entries are of the correct format
- No duplicate entries should be allowed
- All links should take the user to the correct page.

Integration Testing

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

Acceptance Testing

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

6.1.6 Acceptance Testing

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

6.2 Test strategy and approach

Our aim was to uncover maximum number of bugs in minimum amount of time. We succeeded in this objective.

Test objectives

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

Features to be tested

- Verify that the entries are of the correct format.
- No duplicate entries should be allowed.
- All links should take the user to the correct page

Test cases1:

Test case for Login form:

FUNCTION:	LOGIN
EXPECTED RESULTS:	Should Validate the user and check his existence in database
ACTUAL RESULTS:	Validate the user and checking the user against the database
LOW PRIORITY	No
HIGH PRIORITY	Yes

Test case2:**Test case for User Registration form:**

FUNCTION:	USER REGISTRATION
EXPECTED RESULTS:	Should check if all the fields are filled by the user and saving the user to database.
ACTUAL RESULTS:	Checking whether all the fields are field by user or not through validations and saving user.
LOW PRIORITY	No
HIGH PRIORITY	Yes

Test case3:**Test case for Change Password:**

When the old password does not match with the new password ,then this results in displaying an error message as “ OLD PASSWORD DOES NOT MATCH WITH THE NEW PASSWORD”.

FUNCTION:	Change Password
EXPECTED RESULTS:	Should check if old password and new password fields are filled by the user and saving the user to database.
ACTUAL RESULTS:	Checking whether all the fields are field by user or not through validations and saving user.
LOW PRIORITY	No
HIGH PRIORITY	Yes

7.RESULTS

In above screen server started and now open browser and enter URL as <http://127.0.0.1:8000/index.html> and then press enter key to get below page

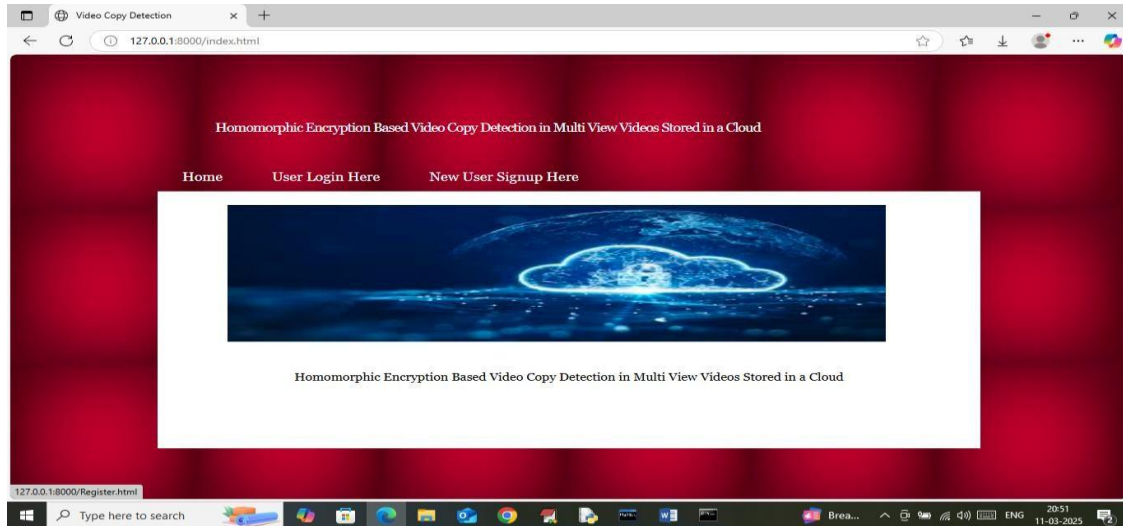


Fig. 7.1 Home page

In above screen click on ‘New User Sign up’ link to get below page

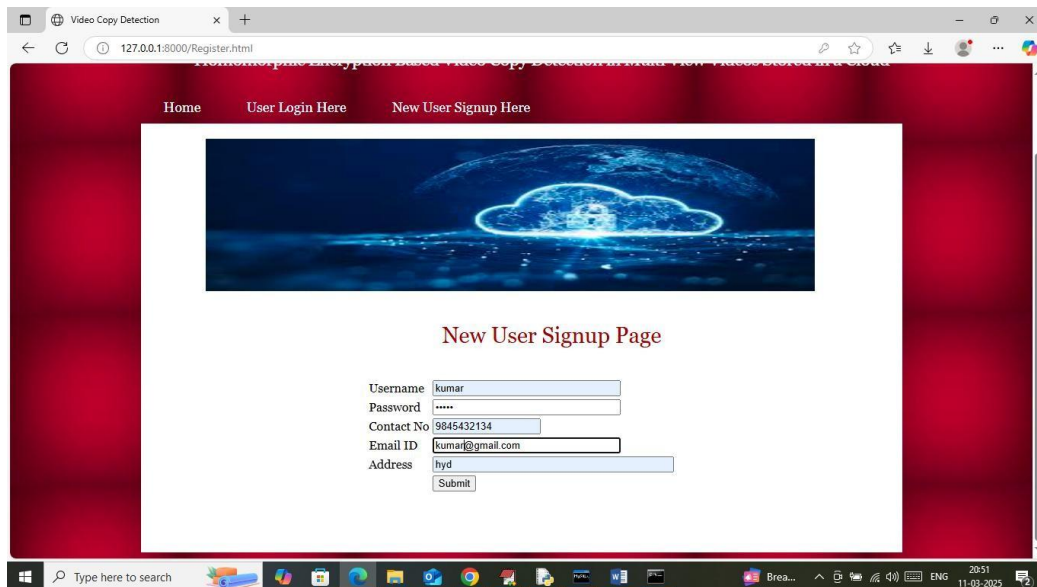


Fig. 7.2 New User Sign up

In above screen user is entering sign up details and then press button to get below page

Video Copy Detection

127.0.0.1:8000/RegisterAction

Home User Login Here New User Signup Here

Homomorphic Encryption Based Video Copy Detection in Multi View Videos Stored in a Cloud

New User Signup Page

Signup process completed. Login to perform video copy detection activities

Username

Password

Contact No

Email ID

Address

Submit

Fig. 7.3 Entering sign up details

In above screen sign up completed and now click on 'User Login' link to get below page

Video Copy Detection

127.0.0.1:8000/UserLogin.html

Home User Login Here New User Signup Here

Homomorphic Encryption Based Video Copy Detection in Multi View Videos Stored in a Cloud

User Login Page

Username

Password

Login

Fig. 7.4 User Login

In above screen user is login and after login will get below page

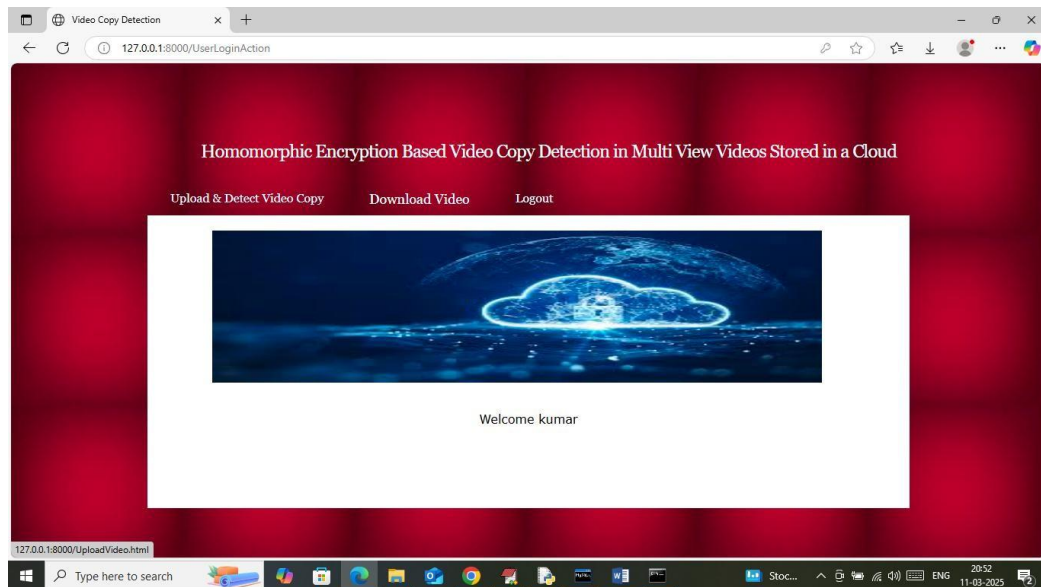


Fig. 7.5 Welcome Page After Login

In above screen user can click on ‘Upload & Detect Video Copy’ link to get below page

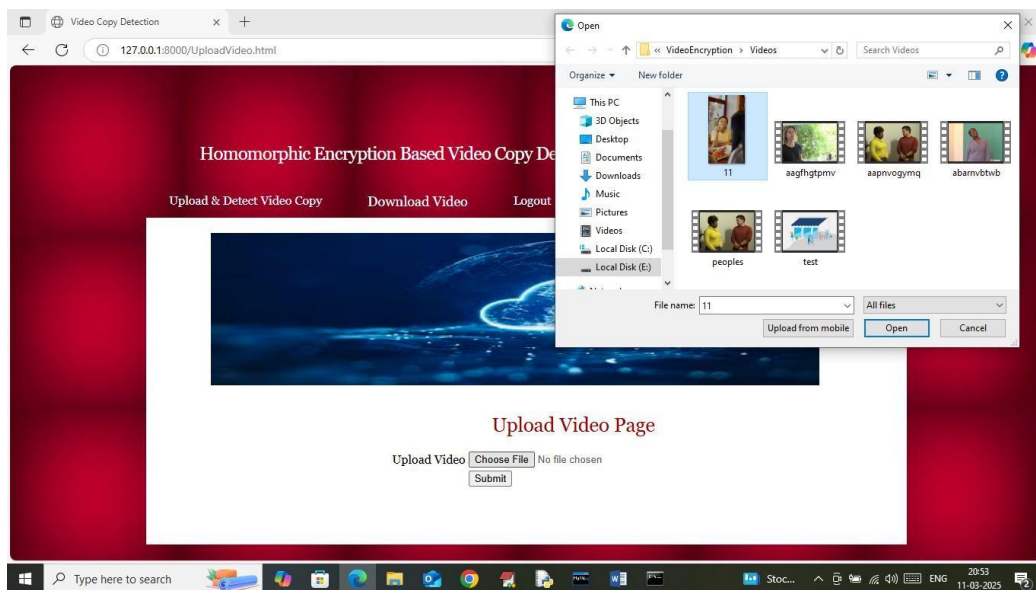


Fig. 7.6 Upload & Detect Video Copy

In above screen selecting and uploading video and then click on buttons to get below page

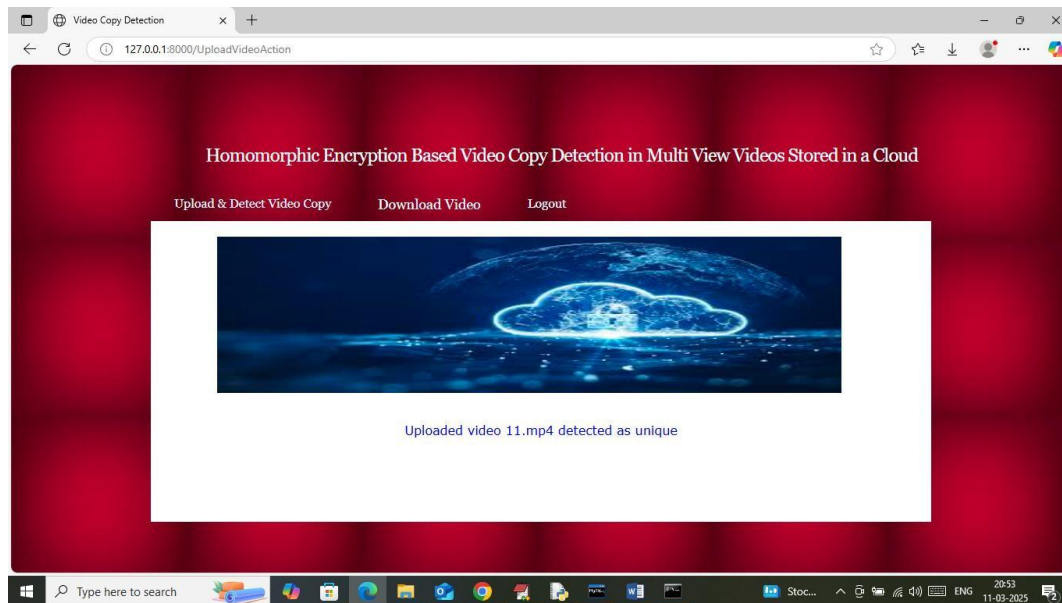


Fig. 7.7 Uploading video

In above screen in blue text can see video detected as 'Unique' and similarly upload some other

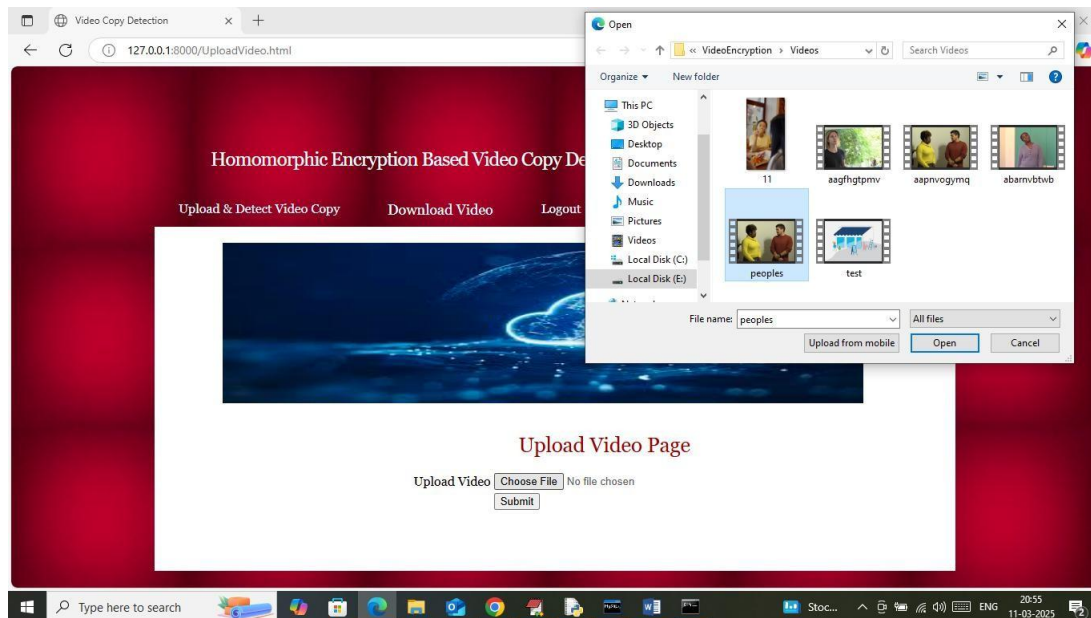


Fig. 7.8 Unique or Copy

In above screen uploading another video and below is the output

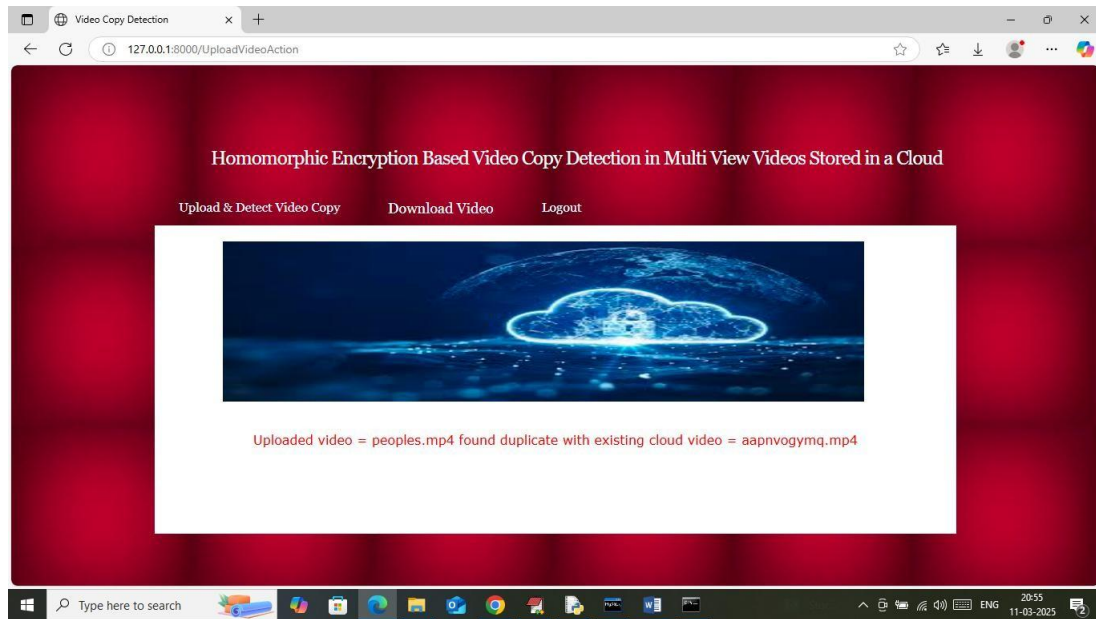


Fig. 7.9 Output

In above screen in red text can see uploaded video matched with cloud video and duplicate found so that video will not be saved and cloud space will be reduced and detection will be possible without decrypting and leaking sensitive data. Now click on 'Download Video' link to get below page

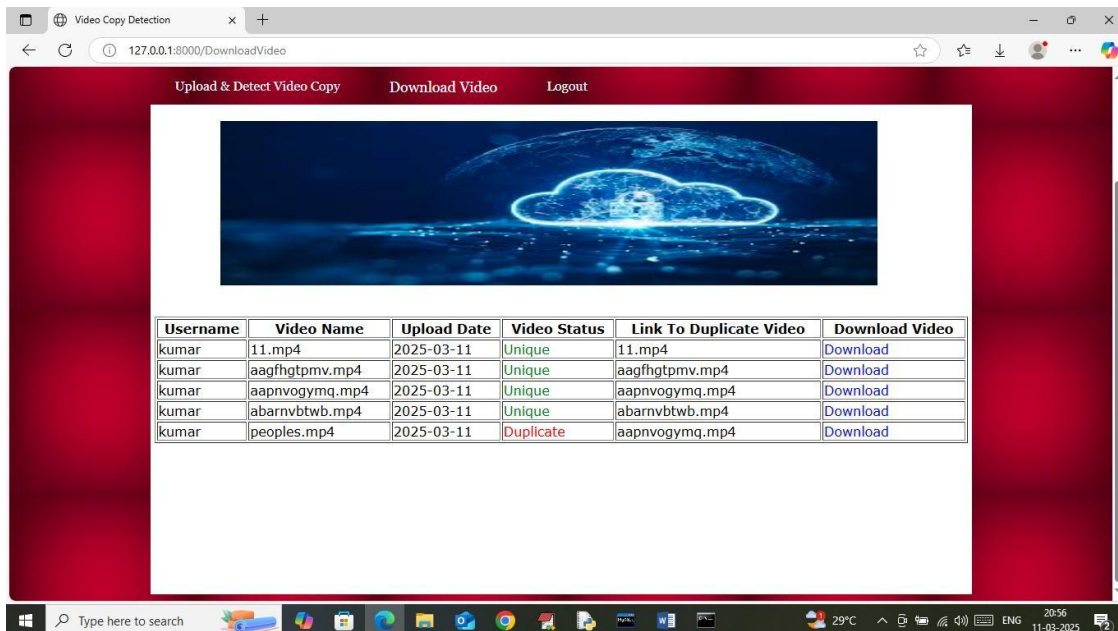


Fig. 7.10 Checking & Downloading Video

8 . CONCLUSION AND FUTURE WORK

CONCLUSION :

This project presents a novel approach to video copy detection by integrating homomorphic encryption with advanced view-invariant feature extraction techniques, tailored for multi-view videos stored in cloud environments. By enabling secure computations directly on encrypted data, the proposed system effectively addresses privacy concerns that traditional copy detection methods fail to resolve. The system ensures that sensitive video content remains confidential throughout the detection process, even when hosted on untrusted cloud platforms.

Furthermore, the adoption of multi-view robust features improves detection accuracy by accommodating variations in camera angles, lighting, and video modifications, making the system highly reliable for real-world applications such as surveillance, media streaming, and copyright protection. The cloud-native architecture facilitates scalability and efficient handling of large video repositories, while the encryption framework maintains data integrity and access control.

Overall, this homomorphic encryption-based framework represents a significant advancement in secure multimedia content management. It not only protects intellectual property rights and user privacy but also complies with emerging data protection regulations. Future work can explore further optimization of homomorphic operations and extend the system to other multimedia formats, contributing to the growing field of privacy-preserving video analytics.

FUTURE WORK:

While the proposed system demonstrates strong capabilities in secure video copy detection, there are several avenues for further improvement and expansion. Future work could focus on optimizing the computational efficiency of homomorphic encryption operations, which currently impose significant overhead on processing times. Research into lightweight or approximate homomorphic encryption schemes may help balance security with real-time performance demands.

Another promising direction is the integration of advanced machine learning models, such as transformer-based architectures or graph neural networks, to enhance the extraction of more discriminative and compact features from multi-view videos. This could improve detection accuracy, especially for heavily edited or manipulated video content.

Expanding the system to support other types of multimedia, such as audio or 3D point cloud data, would broaden its applicability across diverse content types in cloud storage. Additionally, exploring federated learning frameworks combined with homomorphic encryption could enable decentralized, privacy-preserving collaborative copy detection across multiple organizations or cloud providers.

Finally, implementing adaptive thresholding and automated alert prioritization based on confidence scores could improve the system's usability for content owners, helping to reduce false positives and focus resources on the most critical detections. These enhancements would further solidify the system's role as a comprehensive and secure solution for multimedia copyright protection in the evolving digital landscape.

9.REFERENCES

1. Gentry, C. (2009). *A Fully Homomorphic Encryption Scheme*. Stanford University.
2. Brakerski, Z., Fan, J., & Vercauteren, F. (2012). *Fully Homomorphic Encryption without Bootstrapping*. In Advances in Cryptology – ITCC.
3. Cheon, J. H., Kim, A., Kim, M., & Song, Y. (2017). *Homomorphic Encryption for Arithmetic of Approximate Numbers*. In ASIACRYPT.
4. Microsoft SEAL (release 3.6). <https://github.com/microsoft/SEAL>
5. Yu, H., Chen, L., & Wang, L. (2020). *Privacy-Preserving Multimedia Content Copy Detection Based on Homomorphic Encryption*. IEEE Transactions on Information Forensics and Security, 15, 1232-1243.
6. Wang, H., Xu, M., & Han, Q. (2019). *Multi-View Video Copy Detection Using Deep Learning Features*. IEEE Access, 7, 52132-52141.
7. Farid, H., & Popescu, A. (2021). *Cloud-Based Secure Video Copy Detection Using Encrypted Features*. Journal of Information Security and Applications, 57, 102712.
8. Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press.
9. AWS Documentation. (2023). *Security Best Practices for Cloud Storage*. <https://aws.amazon.com/security>
10. Dwork, C., & Roth, A. (2014). *The Algorithmic Foundations of Differential Privacy*. Foundations and Trends® in Theoretical Computer Science, 9(3–4), 211–40.